



OSTİM TEKNİK UNIVERSITY ENGINEERING FACULTY GRADUATION PROJECT REPORT

Intrusion Prevention System (IPS) and Firewall Configuration Using Suricata and pfSense

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Title	Name Surname	Sign
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Abstract

Given the rapid pace of changing digital times of today, network protection mechanisms as pro-active and effective mechanisms have never been as critical as now. The graduation project fulfills the need by suggesting and incorporating a single system of Suricata, a high-performance open-source Intrusion Detection and Prevention System (IDS/IPS), and pfSense, a high-feature and popular open-source firewall operating system. The ultimate objective of the project is to deploy a solid, scalable, and cost-effective security configuration that can properly scan web traffic in real time, identify threats, and block malicious activity actively.

Objectives of the study include the setup and configuration of pfSense in a virtual network environment, Suricata integration on pfSense, and fine-tuning the tools to provide an effective network defense system. The project begins with installing virtual machines which simulate a simple network setup, and subsequently, pfSense WAN and LAN interfaces are configured. Suricata is installed by utilizing pfSense package manager, and it is linked with appropriate network interfaces such that it is able to add monitoring and blocking capability.

Extensive rule sets such as Emerging Threats and Snort community rules are loaded and setup to be optimized for enhanced detection with reduced false positives. Firewall rules in pfSense are setup to be stringent in managing incoming and outgoing traffic based on specified security policies. Numerous different methods of penetration testing are executed through Kali Linux in an effort to test the system's performance and dependability, attacking common weaknesses via scanning, enumeration, and exploitation attempts.

The result shows that how the combination of Suricata with pfSense significantly boosts network security using real-time alerting, logging, and traffic management. The system successfully detected all kinds of intrusion attempts and countered them with timely preventive measures without disrupting valid traffic. Also, its modularity and open-source design make it extremely versatile for a range of applications from educational networks to small and medium enterprises.

In conclusion, the project illustrates the feasibility of employing open-source technologies in the deployment of resilient, scalable, and sustainable network security solutions. Technical feasibility, as well as experience-based deployment, illustrates not only the necessity of continuous rule updates, log auditing, and performance tuning, but also in the future, possible enhancements include integration with SIEM systems for centralized threat data, automation of response strategy, and deployment on large-scale enterprise infrastructures.

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Chapter 1

1. Introduction

During the past ten years alone, the sheer increase in digital technology and web-based services has led to a staggering amount of data being communicated on networks that grows exponentially. While the growth has also enabled unprecedented expansion in communications, commerce, and automation, it has simultaneously come with a broad range of threats and vulnerabilities. Cybersecurity is therefore an organic consideration in the development and deployment of any digital system. Network security is one of the most critical areas of cybersecurity that involves protecting the integrity, confidentiality, and availability of information during transit across networks.

As more advanced and more prevalent cyberattacks arise, traditional security features such as minimal firewalls and anti-virus software are becoming more and more incapable of blocking or detecting sophisticated intrusion attempts. Organizations of all forms and sizes today desire intelligent, proactive, and responsive systems able to monitor network traffic in real-time, detect anomalies, and respond automatically to threats. To meet this need, the increasing installation of Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) has witnessed, and these form an integral part of current cybersecurity architecture.

This graduation project outlines the development and design of a network security system integrating Suricata, an open-source IDS/IPS engine, and pfSense, an open-source router and firewall system. This project aims to design an inexpensive, scalable, and effective solution that is capable of blocking and detecting all manner of cyber attacks, as well as giving administrators insightful information using advanced logging and alerting capabilities.

Through the integration of these two tools, the system will not only actively scan network traffic but also enforce security policies and respond to several lines of attack in real-time. Additionally, because pfSense and Suricata are open-source, this solution is also very accessible and easily modifiable, which makes this solution best for research, laboratory, and small business environments. [9].

1.1. Project Background and Motivation

In the context of the growing number of cybersecurity threats, the justification for this project is providing an affordable yet effective security solution for networks that lack commercial-grade security appliances. Educational institutions, small enterprises, and start-ups are not in a financial position to invest heavily in high-end hardware and propriety software, and therefore are easy targets for attack. This project aims

to bridge that gap by offering a solid alternative based on robust and community-supported open-source tools.

Additionally, the project possesses an educational aspect as it provides a hands-on experience of configuring, deploying, and testing real network defense technology. The project allows for examination of intrusion detection and prevention systems, tuning firewall rules, network monitoring, and use of penetration test methodologies—all essential skills for the cybersecurity field.

1.2. Project Objectives

The objectives of the project are to fulfill technical and educational requirements. The primary objectives are:

- To establish a pfSense and Suricata secure network configuration.
- To install and configure pfSense as a virtual firewall system with set up LAN and WAN interfaces.
- To install and integrate Suricata on pfSense for providing IDS/IPS capabilities.
- To configure rule sets (Emerging Threats, Snort, etc.) to block and detect known malicious patterns.
- To configure and apply firewall rules to filter and control network traffic based on security policies.
- To conduct penetration testing using the tools available on Kali Linux to simulate real attack scenarios.
- To test the performance, detection rate, and system response of the integrated solution.
- To generate reports and logs that provide visibility to the network activity and attempted intrusions.

1.3. Project Scope

The central idea of this project is the installation of an intrusion prevention system based on a virtual lab. The task includes pfSense and Suricata installation and configuration, firewall policy development, and performing penetration testing practice to gauge system performance.

The deployment will be limited to a testbed that simulates normal network conditions. The solution will be tested against typical attack vectors such as port scanning,

brute-force attacks, and exploit payloads. While the system is designed for virtual deployment, the methods used are directly applicable to physical networks.

It must be emphasized that the project is not about specifically developing new security algorithms or protocols and is not within the scope of enterprise-level implementations or integration into leading-edge threat intelligence platforms such as SIEMs (Security Information and Event Management) within the current scope.

1.4. Structure of the Report

The report is structured in a way to provide an unambiguous and uniform sequence of events throughout all phases of the project. It is structured as follows:

- Chapter 1 sets the topic, outlines the motivation, states the objectives, and states the project scope.
- Chapter 2 provides an overview of the theory behind IDS/IPS systems, firewalls, and other security-related cyber defence technologies, including an overview of previous work and relevant technologies.
- Chapter 3 describes the technologies and tools used in the deployment, including pfSense, Suricata, Kali Linux, and the virtualization platform.
- Chapter 4 describes the step-by-step system design, installation, and configuration of the system components.
- Chapter 5 describes advanced configurations, rule tuning, firewall optimizations, and system optimization.
- Chapter 6 tests the system via penetration testing, reviewing the results and discussing system effectiveness.
- Chapter 7 provides an overview of the project outcomes, enumerates the problems faced, and makes suggestions for future implementation and development.

The report is concluded with references and appendices that contain supporting information such as configuration screenshots, network diagrams, test results, and other documents.

Chapter 2

2. Theoretical Background

As the digital transformation alters every facet of society, reliance on interconnected systems has increased exponentially. While such interconnection is ushering in efficiency and innovation, it raises the attack surface for malicious forces. Cyber attacks have become sophisticated and more numerous, making traditional defense mechanisms unable to counter high-level intrusions. In such a scenario, developing intelligent, adaptive, and multi-layered security solutions became the need of the hour.

This chapter aims to put the theoretical basis of the project in place through an examination of fundamental network security concepts and tools like Intrusion Detection and Prevention Systems (IDS/IPS), firewalls, and the open-source products pfSense and Suricata. It also touches on the relevance of penetration testing as a mechanism of testing and validation.

2.1. Introduction to Network Security

Network security is entirely policy, practice, and technology used to secure the usability, integrity, reliability, and security of computer network infrastructure and data. Network security is an essential system defense that deters unauthorized use, misuse, denial, or modification of systems and sensitive data. [13].

Since the arrival of cloud computing, IoT (Internet of Things), distributed workspaces, and increased use of internet-based applications, network landscapes are becoming more dynamic and at risk. Common attacks are:

- Malware attacks, including viruses, worms, and ransomware
- Phishing and social engineering
- Man-in-the-middle (MITM) attacks
- DDoS (Distributed Denial of Service) attacks
- Zero-day exploits
- Privilege escalation and unauthorized access

For overall security, organizations follow the defense-in-depth model that comprises multiple layers of security controls such as endpoint security, encryption, firewalls, and IDS/IPS. Out of all these, two most important technologies are firewalls and IDS/IPS that monitor and control the incoming and outgoing traffic from the network.

2.2. Intrusion Detection and Prevention Systems (IDS/IPS)

Intrusion Detection Systems (IDS) as well as Intrusion Prevention Systems (IPS) are guards that watch and respond to apparently suspicious network traffic with the capability to cause harm to a system. Both are extremely interconnected but distinct entities in a security environment. [13] [15].

- Intrusion Detection System (IDS) watches network traffic and identifies unusual behavior. It is passive in nature and sends alerts to administrators if it identifies any anomaly or familiar attack signature. IDS platforms don't stop traffic but are extremely vital for visibility and forensic analysis.
- Intrusion Prevention System (IPS) is another step in the sense that it operates in inline mode, and hence it has the capability to take active measures such as dropping traffic, dropping connections, or altering firewall policies on detecting threats.

IDS and IPS technologies both can be installed in various detection modes:

- Signature-based detection, where known malicious signatures such as virus definitions or exploit code are used to detect threats.
- Anomaly-based detection, where anomalies are identified from known baseline activity through statistical or machine learning examination.
- Hybrid detection, integrating the two for better coverage and accuracy.

IDS/IPS systems like Suricata can inspect traffic at several OSI model layers. They provide functionalities such as:

- Deep Packet Inspection (DPI)
- Application-layer protocol decoding
- File extraction
- Threat classification and logging

- TLS fingerprinting

Suricata in this project will be configured to operate in IPS mode so that it can block and mark malicious traffic in real-time. This improves network stability and optimizes response time to incidents.

2.3. Overview of Firewall Systems and pfSense

A firewall is an application or device hardware that operates as a perimeter between internal networks known to be trusted and external networks that are not trusted, such as the Internet. A firewall enforces access control policy by filtering traffic arriving and leaving according to an assigned set of rules.

Firewalls are traditionally divided into the following categories:

- Packet-Filtering Firewalls: Examine packets in isolation out of context.
- Stateful Inspection Firewalls: Retain history of active sessions.
- Application-Layer Firewalls: Translate application protocols (for example, HTTP, FTP).
- Next-Generation Firewalls (NGFWs): Include DPI, IPS, antivirus, etc., and other higher-end capabilities.

pfSense is an open-source firewalling and routing software OS based on FreeBSD and popular with enterprise, and home networking use. Its web interface contains extensive features supporting setting up firewalls, VPN, DHCP, DNS, NAT, etc., which can be used by admins for configuration purposes.

- The prime characteristics of pfSense are
- Stateful packet inspection
- Multi-WAN load balancing and failover
- Traffic shaping and bandwidth management
- VPN (IPSec, OpenVPN) support
- Comprehensive logging and monitoring tools

- Integration with packages like Suricata, Snort, Squid

pfSense is utilized here as the foundation for the secure network configuration. Suricata is loaded as a pfSense package and instructed to scan certain interfaces. Through this integration, both firewall and intrusion detection/prevention functionality are centrally managed from a single interface. [1].

2.4. Overview of Suricata IDS/IPS

Suricata is an open-source, high-speed engine that can be used as an IDS, IPS, and Network Security Monitoring (NSM) system by the Open Information Security Foundation (OISF). Suricata is world-famous for its multi-threading capabilities and high-performance design and supports large-scale logging, deep packet inspection, and real-time traffic analysis.

Suricata provides:

- Protocol detection and parsing (e.g., HTTP, TLS, SSH, FTP)
- Signature-based detection with Emerging Threats and Snort rules
- Inline prevention capabilities
- Rich logging in EVE, JSON, and syslog
- File extraction, TLS certificate analysis, and geolP searches

Suricata rule engine supports a dynamic syntax, with which security analysts can indicate certain patterns or behavior that are indicative of malicious activity. Suricata here is running in inline IPS mode on the LAN interface, by virtue of which it can observe as well as drop unauthorized or malicious connections.

Suricata's alerting capability provides real-time information on network attacks, making it most appropriate for inclusion with SIEMs and security panels. [2].

2.5. Penetration Testing and Kali Linux

Penetration testing, or ethical hacking, is a method used to evaluate the security of a system by simulating real attack situations. It is an effective method to detect vulnerabilities, test defenses, and measure system stability.

Kali Linux is a Debian-based Linux distribution widely utilized by security professionals and ethical hackers. It is preinstalled with hundreds of tools for:

- Network scanning (e.g., Nmap, Netdiscover)
- Vulnerability exploitation (e.g., Metasploit, ExploitDB)
- Password attacks (e.g., Hydra, John the Ripper)
- Packet sniffing and analysis (e.g., Wireshark)
- Social engineering, privilege escalation, and more

Kali Linux is used in this project to simulate various kinds of attacks—port scanning, brute-force login, and payload injection—in an attempt to stress the ability of the Suricata-pfSense system to detect and block threats. Such tests are used to validate rule sets and firewall configurations' effectiveness in practical use.

Chapter 3

3. Tools and Technologies

This chapter provides an in-depth description of the technologies, platforms, and tools utilized in the network security system development, configuration, and testing of this project. Those technologies were chosen for their reliability, open-source platform, and suitability for experimental and academic network environments. Each of the components played diverse roles in the system, including intrusion prevention and firewall configuration through to attack simulation and testing.

3.1. pfSense

pfSense is a free open-source firewall and router operating system with origins in FreeBSD. It is highly stable, highly scalable, and easy to set up with a web-based graphical user interface. pfSense has a comprehensive set of features found in expensive commercial firewall boxes such as packet filtering, VPN, NAT, DNS forwarding, VLAN management, bandwidth limiting, and system monitoring. [1].

In the current project, pfSense is used as the main firewall and routing platform. It is deployed in a virtualized environment and configured with WAN and LAN interfaces for the sake of emulating real-world networking environments. Through its Package Manager, pfSense includes the functionality of installing extra security software such as Suricata, which was required for the incorporation of IDS/IPS functionality within the same interface. This combination provides a consolidated and dynamic platform for the management of network traffic, setting up access rules, and inspecting packets for malicious behavior.

- Key pfSense features utilized in this project:
- Web-based admin console
- Stateful packet inspection and rule-based filtering
- DHCP server and DNS resolver
- NAT and port forwarding
- Package installer for the installation of Suricata

3.2. Suricata

Suricata is an open-source high-speed IDS/IPS and Network Security Monitoring (NSM) engine developed by the Open Information Security Foundation (OISF).

Suricata provides various operating modes like passive detection, inline prevention, and traffic analysis owing to its support for multi-threading and deep packet inspection.

Suricata employs user-provided rule sets such as Emerging Threats (ET) and Snort rules in order to identify known malicious signatures and act on them. Suricata is able to inspect protocols such as HTTP, TLS, FTP, and DNS and generate formatted alerts and logs per incident. Suricata also provides support for file extraction, TLS fingerprinting, and flow logging, thereby making it an end-to-end solution for advanced network security monitoring. [2]

Suricata is installed using pfSense package manager in this project itself and is set up to sniff LAN interface. It is set to start in inline IPS mode so that it can block the evil traffic in real time using pre-configured sets of rules. Alert and log options also provide end-to-end visibility into any system activity and threats.

- Suricata tasks in this project are primarily:
- Real-time intrusion detection and prevention
- Application layer protocol inspection
- pfSense integration for remote management
- Port scanning, brute-force, and exploit payload detection

3.3. Kali Linux

Kali Linux is a Debian Linux distribution maintained and developed by Offensive Security particularly for penetration testing, digital forensic analysis, and ethical hacking. It contains over 600 security tools pre-installed and is among the most widely used and advanced operating platforms for testing the security strength of computer systems.

In this project, the Kali Linux operating system is used to perform virtual cyberattacks on the pfSense + Suricata configuration. The attacks need to be performed in order to confirm the effectiveness of IDS/IPS and firewall rules. The following tools were used during the penetration testing:

- Nmap – for network and port scanning
- Metasploit Framework – for exploiting discovered vulnerabilities and inserting payloads

Utilizing Kali Linux allows one to possess a real and controlled test lab through which the defense mechanisms of the system can be tested under various attack approaches. [4]

3.4. Virtualization Platform: VMware / VirtualBox

To test and deploy the network configuration in a safe and isolated environment, the project employs a virtualization platform such as Oracle VM VirtualBox or VMware Workstation. These platforms allow one to create several virtual machines (VMs), each simulating different roles such as:

- The pfSense firewall/router
- A client device behind the firewall
- An attacker machine on Kali Linux

Using virtualization has with it a number of significant benefits:

- Hardware independence: There is no requirement of physical routers, firewalls, or additional computers
- Snapshot and rollback facility: Can be employed for restoring the system at configuration failures
- Scalability: Easy addition of additional virtual nodes in the test network
- Network simulation: LAN/WAN bridging, NAT, and isolated internal networks are handled in a flexible manner

This arrangement supports repeated testing, configuration optimization, and quick prototyping with zero risk to physical network infrastructure.

3.5. Other Tools and Resources

Apart from the main technologies applied in firewalling, intrusion detection, and penetration testing, there were other tools and resources available to facilitate the analysis, test, and documentation phases of this project. These tools were utilized significantly in packet analysis, vulnerability scanning, data visualization, and rule management.

- Wireshark

Wireshark is a protocol network analyzer employed extensively to analyze live and captured network traffic in minute detail. Wireshark was used in the project to investigate packet behavior, try Suricata rule triggering, and verify that Suricata and pfSense reacted as anticipated to the test attacks.

- Nessus Vulnerability Scanner

Nessus by Tenable is a widely employed vulnerability scanner utilized to scan systems and networks for publicly known security vulnerabilities. Nessus was used in this project to perform detailed vulnerability scans on behind-the-pfSense-firewall systems. This provided comprehensive reports of system misconfigurations, old services, and vulnerabilities open to known CVEs. These were later used to verify whether Suricata detected the scan attempts and the efficiency of pfSense in restricting access according to established firewall rules.

The inclusion of Nessus added to the pen testing done manually via Kali Linux by ensuring a methodical and automated system for vulnerability scanning. It also allowed for a check of system readiness against passive but equally malicious reconnaissance attacks.

Chapter 4

4. System Design and Implementation

This chapter describes the whole technical setup of the network security system, focusing on the setup of the virtual environment, pfSense installation and configuration, integration of Suricata, and implementation of firewall and intrusion prevention rules. The steps were implemented in a virtual lab using pfSense as the firewall platform and Suricata as the IDS/IPS engine. Screenshots of every step of the implementation process are provided to visually document the system development.

4.1. Network Topology Design

In order to simulate a standalone, safe network, VirtualBox or VMware Workstation was utilized to build a virtual lab environment. The setup consisted of the following virtual machines:

- pfSense VM: Serving as the router/firewall
- Kali Linux VM: Serving as the attacking machine
- Windows/Linux Client VM: For simulating a user device within the internal

The pfSense VM had two virtual network interfaces assigned to it:

- WAN Interface: Connected to a bridged adapter or NAT (outside access)
- LAN Interface: Connected to the internal network

This mimicked a basic real-world setup where the firewall monitors and manages traffic between the trusted and untrusted regions.

4.2. pfSense Installation

The pfSense ISO image used was downloaded and installed onto a virtual machine with recommended system resources. After the first boot and console setup, the web-based GUI (WebConfigurator) was accessed via the LAN IP (e.g., <https://192.168.1.1>).

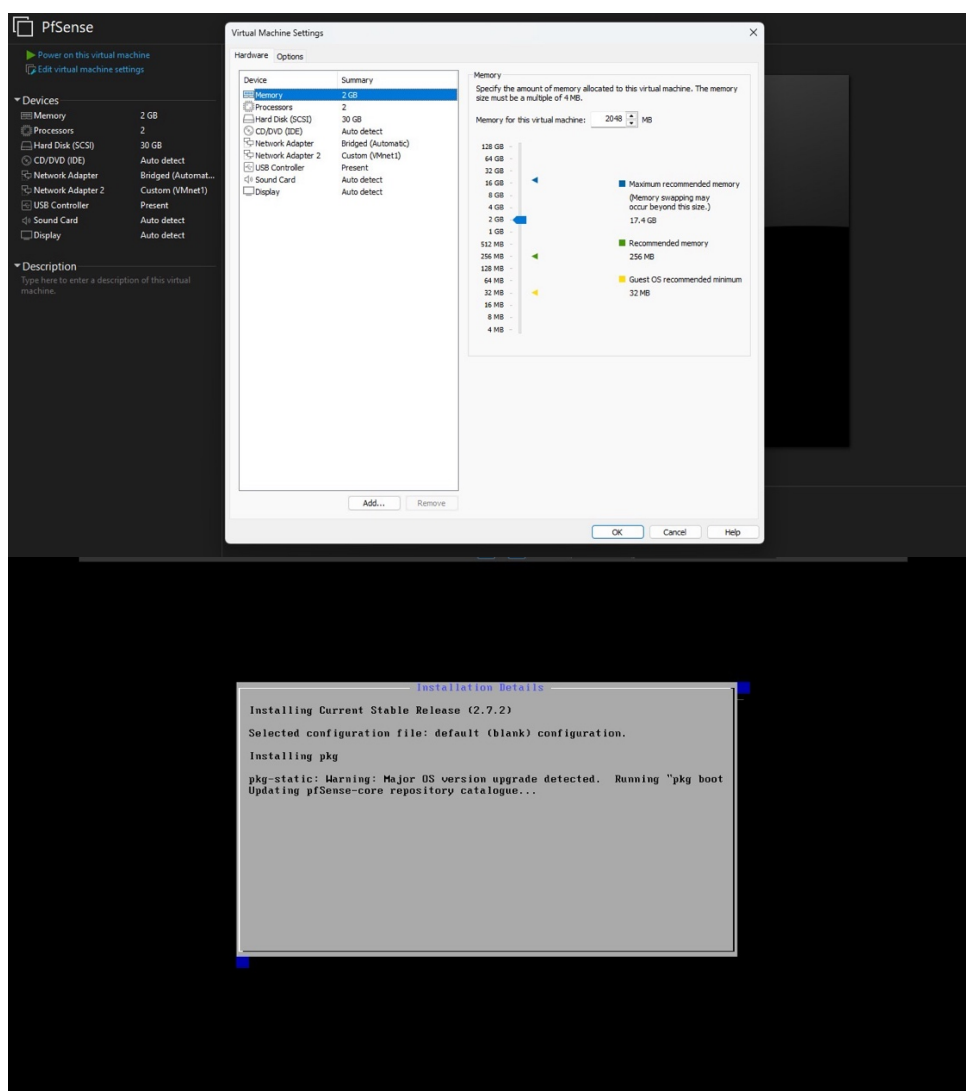
The pfSense setup wizard provided simple setting up of:

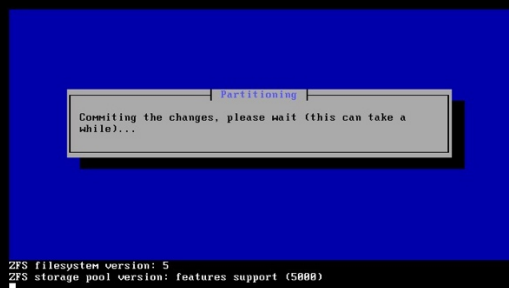
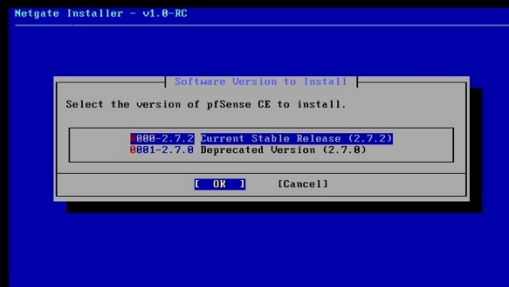
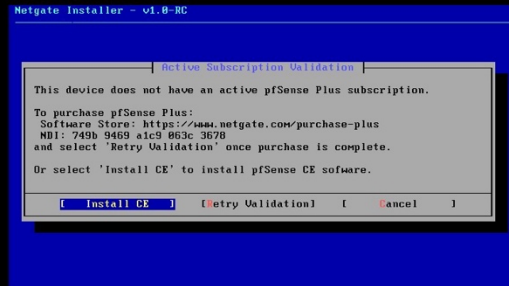
- Hostname, DNS, and gateway

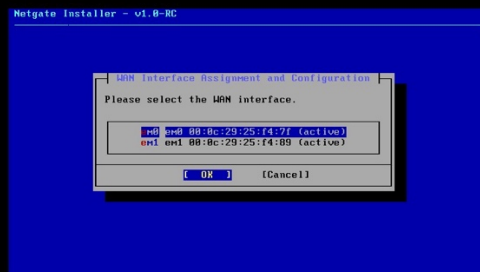
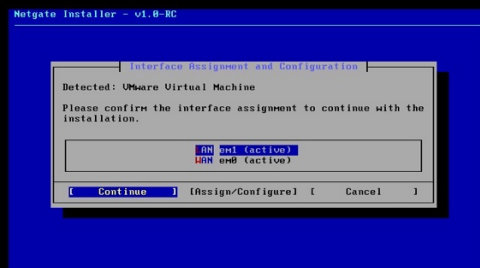
- WAN and LAN IP settings
- Admin password
- DHCP and DNS services

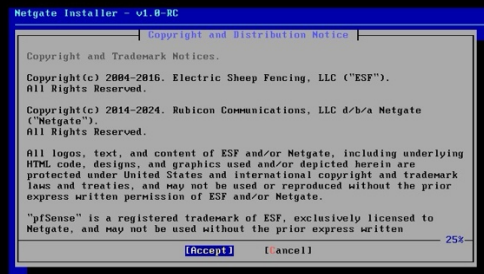
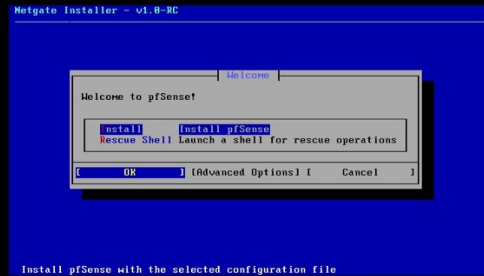
Upon completion, pfSense presented a responsive and stable interface to configure firewall rules and install extra packages.

And the following snapshots from the pfSense Installation:









```
Starting CRON... done.
Starting package suricata...done.
pfSense 2.7.2-RELEASE amd64 28248384-1953
Bootup complete

FreeBSD/amd64 (pfSense.Graduation.lab) (ttyv0)
VMware Virtual Machine - Netgate Device ID: 4e5778fddd60be51a33c

*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

LBN (wan) -> em0 -> v4/BHCP4: 192.168.1.183/24
LBN (lan) -> em1 -> v4: 192.168.1.1/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell - pfSense tools
4) Reset to factory defaults 13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                15) Restore recent configuration
7) Ping host                  16) Restart PHP-FPM
8) Shell

Enter an option: █
```

Activities Firefox Web Browser

pfSense - Login

https://192.168.1.1

pfSense

Login to pfSense

SIGN IN

Username

Password

SIGN IN

pfSense is developed and maintained by Netgate. © ESF 2004 - 2025 View license.

pfSense Graduation Lab

https://192.168.1.1/pkg_mgr_installed.php

pfSense COMMUNITY EDITION

System / Package Manager / Installed Packages

Installed Packages Available Packages

Name	Category	Version	Description	Actions
✓ suricata	security	7.0.8_1	High Performance Network IDS, IPS and Security Monitoring engine by OISF. Package Dependencies: suricata-7.0.8	Update ☒ Current Remove Information Reinstall

Newer version available

Package is configured but not (fully) installed or deprecated

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pfSense Graduation Lab

https://192.168.1.1

pfSense COMMUNITY EDITION

Status / Dashboard

System Information	
Name	pfSense.Graduation.lab
User	admin@192.168.1.101 (Local Database)
System	VMware Virtual Machine Netgate Device ID: 4e5770fddd60be51a33c
BIOS	Vendor: Phoenix Technologies LTD Version: 6.00 Release Date: Thu Nov 12 2020
Version	2.7.2-RELEASE (amd64) built on Mon Mar 4 22:53:00 +03 2024 FreeBSD 14.0-CURRENT The system is on the latest version. Version information updated at Tue Mar 18 17:46:27 +03 2025
CPU Type	Intel(R) Core(TM) i7-8550U CPU @ 1.80GHz 2 CPU(s): 2 package(s) x 1 core(s) AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No
Hardware crypto	Inactive
Kernel PTI	Enabled

Netgate Services And Support	
Contract type	Community Support Community Support Only
NETGATE AND pfSense COMMUNITY SUPPORT RESOURCES	
If you purchased your pfSense gateway firewall appliance from Netgate and elected Community Support at the point of sale or installed pfSense on your own hardware, you have access to various community support resources. This includes the NETGATE RESOURCE LIBRARY . You also may upgrade to a Netgate Global Technical Assistance Center (TAC) Support subscription. We're always on! Our team is staffed 24x7x365 and committed to delivering enterprise-class, worldwide support at a price point that is more than competitive when compared to others in our space.	
- Upgrade Your Support - Netgate Global Support FAQ - Netgate Professional Services	- Community Support Resources - Official pfSense Training by Netgate - Visit Netgate.com
If you decide to purchase a Netgate Global TAC Support subscription, you MUST have your Netgate Device ID (NDI) from your firewall in order to	

4.3. Suricata Installation and Configuration

After pfSense was installed and configured properly, the subsequent activity was integrating Suricata to provide intrusion detection and prevention capabilities. Suricata was installed through the pfSense package management system directly, which ensured complete compatibility with the firewall platform.

The installation was done by executing the following steps:

- Logged into the pfSense web interface and proceeded to System → Package Manager → Available Packages.
- Searched and installed Suricata from within packages.
- Invoked and completed the installation process by using the pfSense Package Manager.

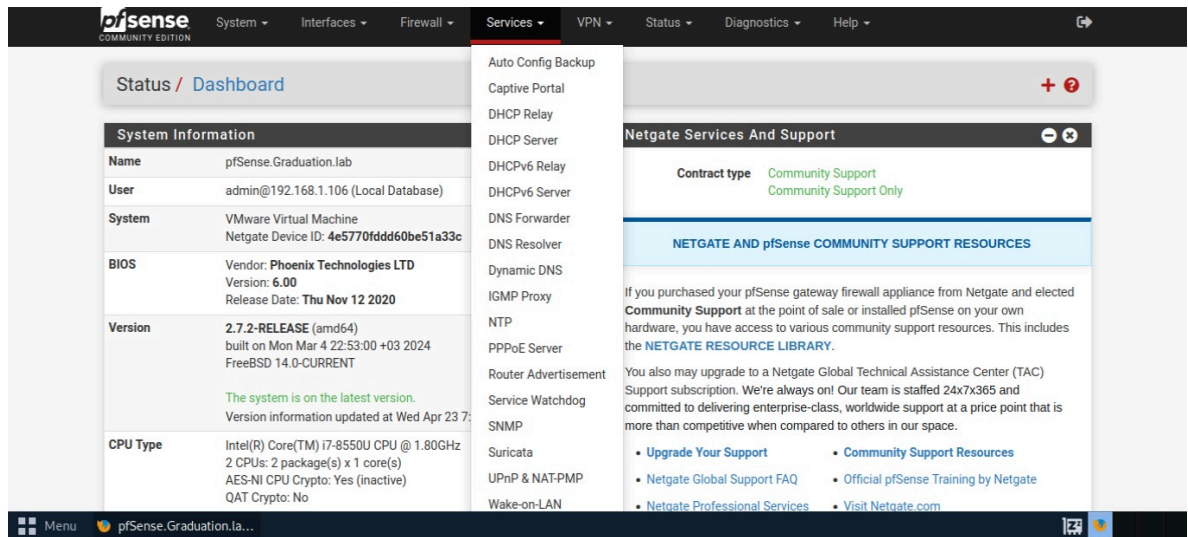
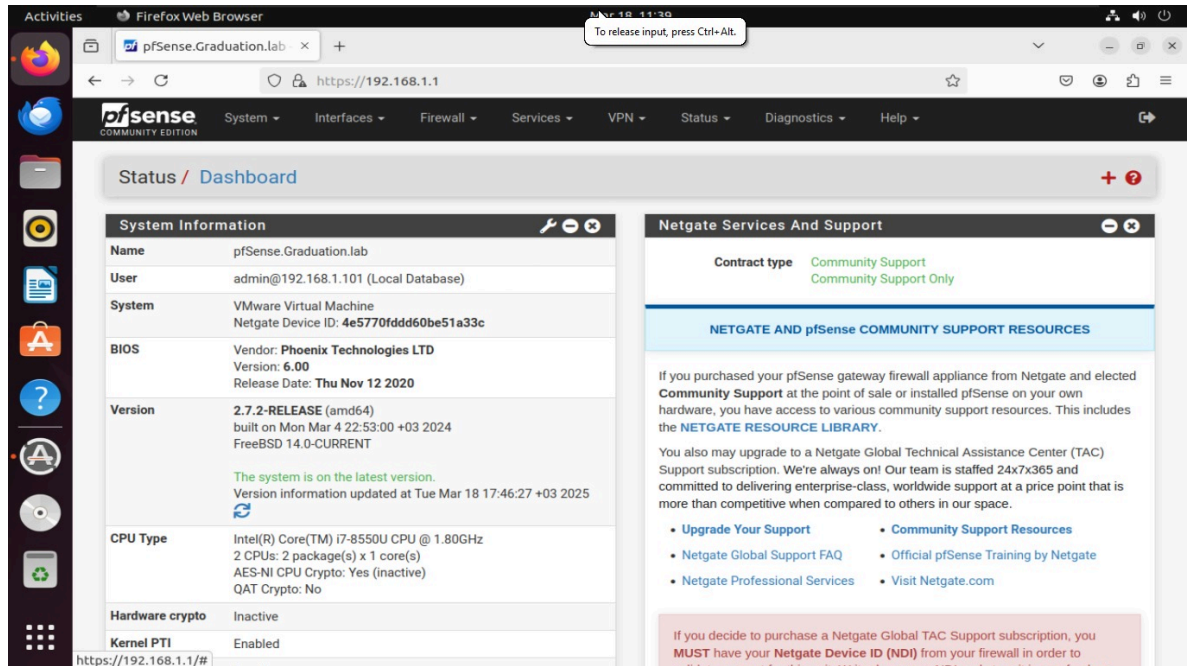
Upon successful installation, Suricata was accessible via Services → Suricata in the pfSense web interface. The setup process involved:

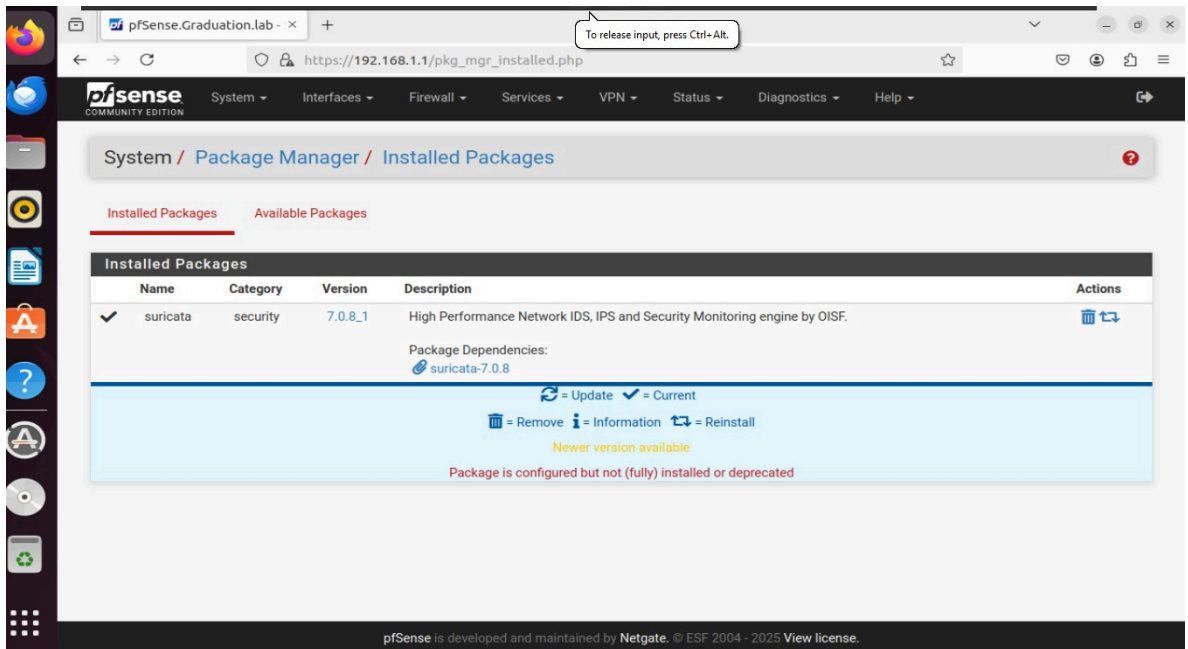
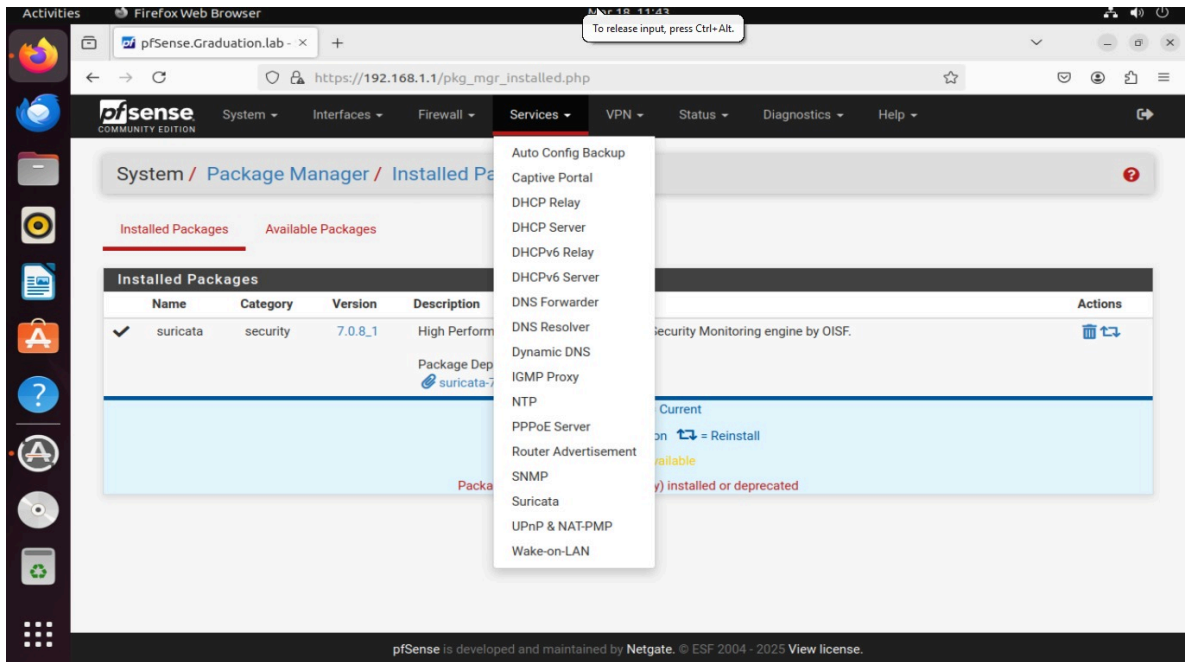
- Interface Assignment:
Suricata was configured to listen on the LAN interface so that the system could scan and protect internal network traffic effectively.
- Enabling Inline IPS Mode
The inline mode was enabled so Suricata not only could detect threats but actively block malicious traffic in real time, essentially acting as an Intrusion Prevention System (IPS).
- Rule Set Management:
Multiple rule sets were selected and downloaded, including:
 - I. Emerging Threats (ET Open): Community-open-source managed detection rule set at no charge.
 - II. Snort GPLv2 Community Rules: Rules written by Snort, Suricata-compatible that add detection capability.
- Suricata was also configured to run automatically upon system startup for constant protection.

Moreover, alert logging was done through the EVE JSON format for creating parsable and structured logs that can be analyzed later.

This basic Suricata configuration placed the system to perform real-time examination, detection, and blocking of malicious traffic as part of an integrated layer defense of the network. [6]

Screenshots demonstrating the installation steps are provided here:





pfSense
COMMUNITY EDITION

System Interfaces Firewall Services VPN Status Diagnostics Help

Services / Service Watchdog

This page allows selecting services to be monitored so that they may be automatically restarted if they crash or are stopped.

Monitored Services

Notify	Service Name	Description	Actions
☐	suricata	Suricata IDS/IPS Daemon	

+ Add New Service

Save Notification Settings

Delete

Check Notify next to services to perform an e-mail notification when the service is restarted. Configure e-mail notifications to receive the alerts.

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Menu pfSense.Graduation.la...

pfSense
COMMUNITY EDITION

System Interfaces Firewall Services VPN Status Diagnostics Help

Services / Suricata / Updates

Interfaces Global Settings Updates Alerts Blocks Files Pass Lists Suppress Logs View Logs Mgmt SID Mgmt

Sync IP Lists

INSTALLED RULE SET MD5 SIGNATURES

Rule Set Name/Publisher	MD5 Signature Hash	MD5 Signature Date
Emerging Threats Open Rules	c63abd60bf09e1e56e383dfe4109ea72	Wednesday, 23-Apr-25 08:10:17 +03
Snort Subscriber Rules	Not Enabled	Not Enabled
Snort GPLv2 Community Rules	bb8443a49603820bb1e2528161e88d25	Wednesday, 23-Apr-25 08:10:17 +03
Feodo Tracker Botnet C2 IP Rules	3c6031ba2ac14b01acb1e855c1d6d2c1	Wednesday, 23-Apr-25 08:10:17 +03
ABUSE.ch SSL Blacklist Rules	8feee62c439d6e7207fd970bc21f9fe	Wednesday, 23-Apr-25 08:10:17 +03

UPDATE YOUR RULE SET

Last Update: Apr-23 2025 08:10

Result: success

Menu Parrot Terminal pfSense.Graduation.la...

Chapter 5

5. Configuration and Optimization

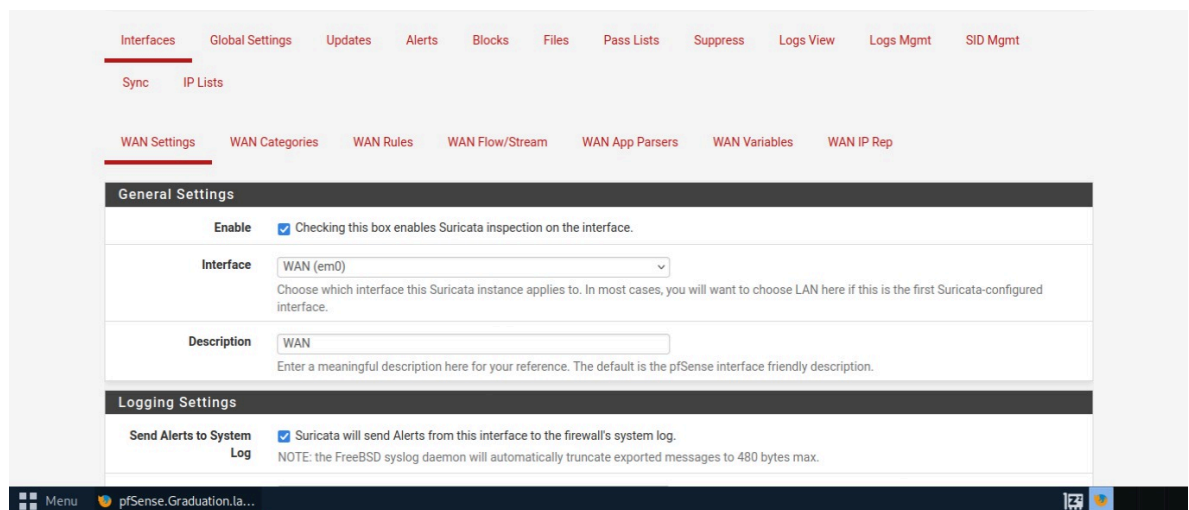
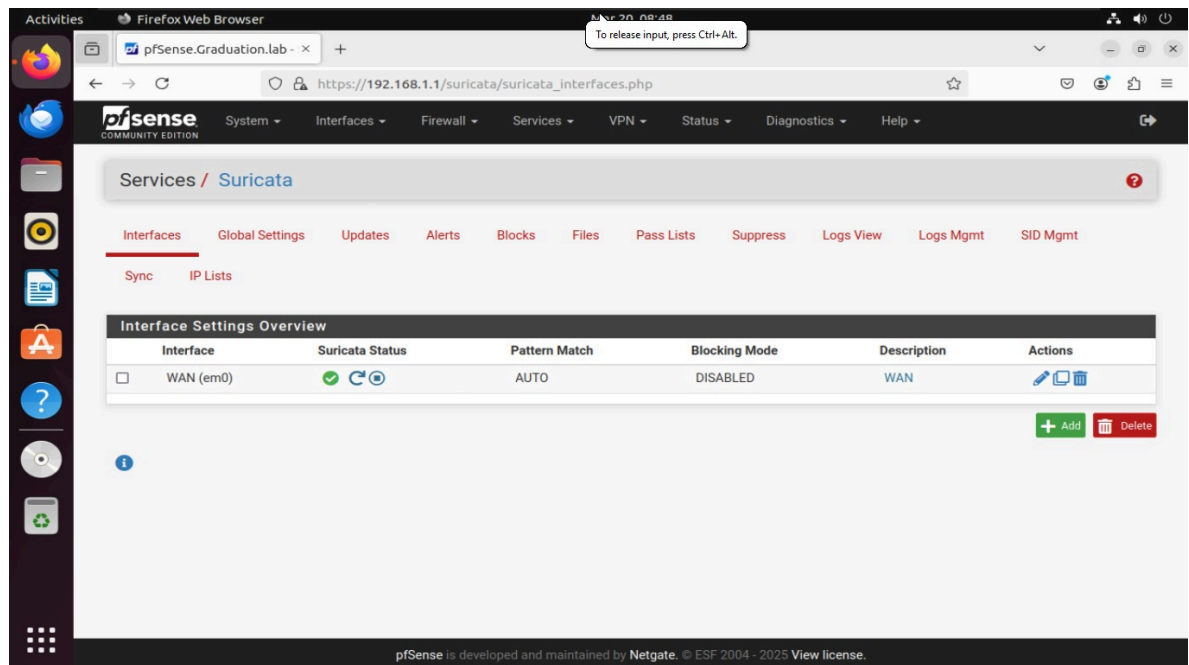
After the main installation and initial configuration of pfSense and Suricata, further configurations and optimizations were also conducted to further enhance the optimization of the system's performance, reliability, and responsiveness. This chapter presents the most critical configuration methods utilized to guarantee appropriate functionality of the intrusion detection and prevention system under various network conditions, with specific focus on minimization of false positives and cost of performance.

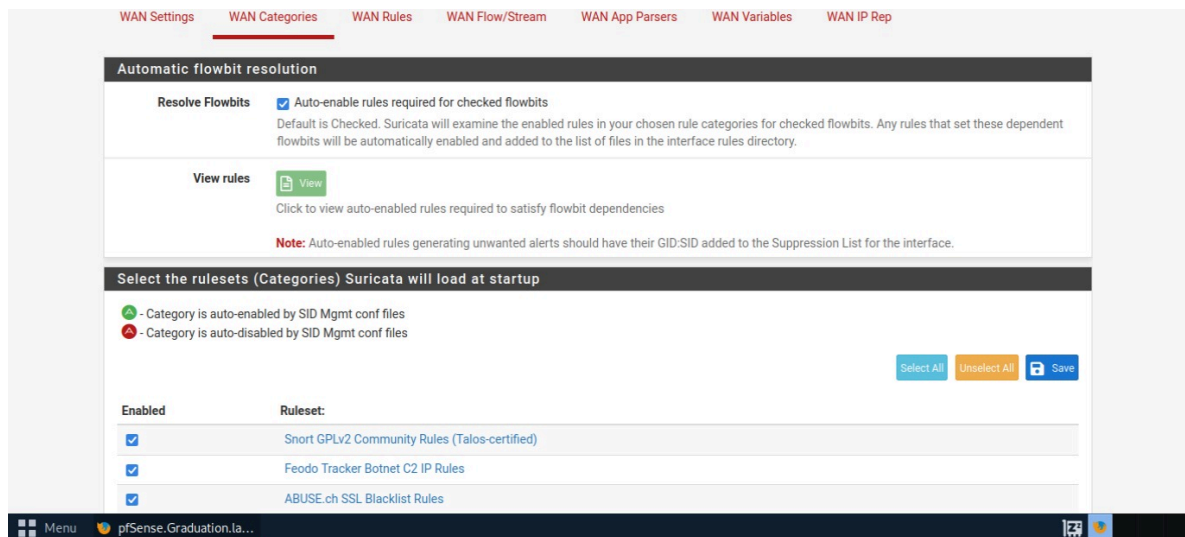
5.1. IDS/IPS Mode Activation and Rule Management

Once Suricata was properly bound to the LAN interface, its operating mode was configured with caution to ensure optimal detection and prevention capabilities.

- Inline Mode Activation:
- Suricata was configured specifically to operate in inline IPS mode. Under inline mode, Suricata actively scans and enforces traffic decisions by dropping or rejecting packets that carry threat signatures, instead of just alerting administrators.
- Rule Set Updates and Management:
- Emerging Threats (ET Open) and Snort GPL rule sets were enabled. Periodic automatic updates were configured to ensure that detection signatures remain current against the latest exploits and attack techniques.
- Selective rule categories were turned on based on relevance to the network environment, such as:
 - a. Malware communications
 - b. Web application attacks
 - c. SSH and FTP brute-force attacks
 - d. DDoS patterns
 - e. Port scanning activity
- Thresholding and Suppression:
For the sake of reducing false positives, suppression and thresholding features were activated on certain noisy signatures, primarily for valid internal traffic such as DNS queries and HTTPS connections. [6] [7]

Screenshots demonstrating the configuration steps are provided here:





5.2. Logging, Alerting, and Notifications

Logging and alerting correctly are essential components of any security monitoring configuration. [7] [14]

- **Alert Logging:**
Suricata was configured to log alerts as EVE JSON format so that reporting would be structured and specific. The logs captured source IP and destination IP addresses, protocol information, time stamp, and the related signature ID that was matched for each of the events being monitored.
- **Dashboard Integration:**
Suricata alerts were exposed via the pfSense web GUI. This allowed immediate viewing of active threats without the need for any additional tools.
- **Email Notifications:**
Optional email notifications were supported for severe alerts so administrators could be notified immediately of high-severity occurrences even when they were not logged in to the system.
- **Syslog Forwarding:**
(Optional if configured) Suricata alerts can also be directed to a remote syslog server or SIEM appliance for centralized logging and correlation as part of a larger security configuration.

Screenshots demonstrating the configuration steps are provided here:

Alert and Block Settings

Block Offenders ☒ Checking this option will automatically block hosts that generate a Suricata alert.

IPS Mode Legacy Mode

Select blocking mode operation. Legacy Mode inspects copies of packets while Inline Mode inserts the Suricata inspection engine into the network stack between the NIC and the OS. Default is Legacy Mode.

Legacy Mode uses the PCAP engine to generate copies of packets for inspection as they traverse the interface. Some "leakage" of packets will occur before Suricata can determine if the traffic matches a rule and should be blocked. Inline mode instead intercepts and inspects packets before they are handed off to the host network stack for further processing. Packets matching DROP rules are simply discarded (dropped) and not passed to the host network stack. No leakage of packets occurs with Inline Mode. WARNING: Inline Mode only works with NIC drivers which properly support Netmap! Supported drivers include: bnxt, cc, cxgbe, cxl, em, ena, ice, igb, igc, ix, ixgbe, ixl, lem, re, vmx, vtnet. If problems are experienced with Inline Mode, switch to Legacy Mode instead.

Kill States ☒ Checking this option will kill firewall states for the blocked IP. Default is Checked.

Which IP to Block BOTH

Select which IP extracted from the packet you wish to block. Choosing BOTH is suggested, and it is the default value.

Block On DROP Only ☐ Checking this option will insert blocks only when rule signatures having the DROP action are triggered. When not checked, any rule action (ALERT or DROP) will generate a block of the offending host. Default is Not Checked.

IP Pass List default [View List](#)

Choose the Pass List you want this interface to use. Addresses in a Pass List are never blocked. Select "none" to prevent use of a Pass List.

The default Pass List adds Gateways, DNS servers, locally-attached networks, the WAN IP, VPNs and VIPs. Create a Pass List with an alias to...

Services / Suricata / Blocked Hosts

Interfaces Global Settings Updates Alerts **Blocks** Files Pass Lists Suppress Logs View Logs Mgmt SID Mgmt

Sync IP Lists

Blocked Hosts Log View Settings

Save or Remove Hosts [Download](#) [Clear](#)

All blocked hosts will be saved All blocked hosts will be cleared

Save Settings [Save](#) ☒ Refresh

Save auto-refresh and view settings Default is ON Number of blocked entries to view. Default is 500

Last 500 Hosts Blocked by Suricata

Note: Only blocked IP addresses from Legacy Mode interfaces are shown! For inline IPS mode interfaces, dropped IP addresses are highlighted on the ALERTS tab.

Blocked IP	Block Date/Time	Block Alert Description	Block Rule GUID:SID	Remove Block
172.66.44.77	04/23/2025 07:18:30	ET INFO Observed Cloudflare Page Developer Domain (pages .dev in TLS SNI)	1:2057746	X
	04/10/2025 11:28:09	ET INFO Observed Cloudflare Page Developer Domain (pages .dev in TLS SNI)	1:2057746	

5.3. Configuration of Firewall Rules

To ensure a secure and controlled network traffic flow, firewall rules were carefully planned and configured in pfSense. The rules supplemented Suricata's intrusion prevention capability to provide a layered defense mechanism, striking a balance between accessibility for legitimate users and strict control over potentially malicious connections. [10]

LAN Rules

On the LAN interface, the following rules were configured to regulate outbound traffic:

- **Anti-Lockout Rule:**
This built-in rule ensures that administrators can always access the pfSense WebConfigurator (HTTPS on port 443) from the LAN. It prevents accidental lockouts during firewall rule misconfigurations.
- **Default Allow LAN to Any Rule (IPv4 and IPv6):**
Initially, broad rules were enabled to allow all LAN traffic to any destination. These rules were kept temporarily during initial setup phases but were later tightened with more specific traffic allowances.
- **Allowing Secure Web Access (HTTPS - Port 443):**
A specific rule was created to permit web browsing over encrypted HTTPS connections, ensuring secure communication with external websites.
- **Allowing Internet Browsing (HTTP - Port 80):**
A rule was added to permit standard HTTP traffic, allowing access to websites that may not support HTTPS. Although less secure, this was necessary for compatibility with certain services.
- **Allowing Domain Name Service (DNS - Port 53):**
DNS traffic was allowed so that LAN clients could resolve domain names to IP addresses, essential for normal web browsing and application functionality.
- **Allowing File Transfer Protocol (FTP - Port 21):**
FTP access was permitted to enable file transfer operations with external servers, a service sometimes required for downloading updates or files.
- **Allowing Email Sending (SMTP - Port 25):**
Outbound email sending was enabled by allowing SMTP traffic from LAN clients to external mail servers.
- **Allowing Email Receiving (POP3 - Port 110):**
A rule was created to permit LAN clients to retrieve emails using the POP3 protocol.
- **Allowing Secure Shell (SSH - Port 22):**
SSH traffic was allowed for secure remote access to external servers, facilitating administrative tasks and secure data transfer.
- **Allowing Ping (ICMP):**
ICMP traffic (Ping requests and responses) was permitted for network diagnostics and troubleshooting purposes.

Each LAN rule was configured to log permitted traffic, ensuring visibility into all network communications originating from the internal network.

WAN Rules

On the WAN interface, a default security policy was enforced to block unauthorized and potentially malicious incoming traffic. Two important pre-configured rules were applied:

- **Block Private Networks:**

This rule blocks incoming traffic originating from RFC1918 private IP address spaces (e.g., 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16) on the WAN interface. Such addresses should not appear on the public internet and may indicate spoofing or misconfigured devices.

- **Block Bogon Networks:**

This rule blocks incoming traffic from bogon IP addresses, which are ranges not officially assigned by IANA (Internet Assigned Numbers Authority). These addresses are often used in malicious activities or by misconfigured networks.

By blocking both private and bogon networks on the WAN side, pfSense protects the internal LAN against common scanning, spoofing, and reconnaissance attacks.

All other incoming connections are denied unless explicitly allowed by additional firewall rules.

Floating Rules

Floating rules were employed in order to insert more generic security policies that had an effect across multiple interfaces simultaneously. These comprised:

- Blocking well-known malicious IP addresses detected by Suricata alerts.
- Rate limiting suspicious traffic activity such as ICMP floods or repeated connection attempts.
- Inserting additional logging to follow specific events to be examined in greater detail.

Floating rules were more versatile and inserted general control over traffic flow direction, both encompassing interface-specific and global firewall policies.

Rule Logging and Priority

All critical firewall rules were configured to log for tracking. The rule precedence was tightly controlled based on pfSense's top-down processing design to guarantee that:

- Deny rules were tested before allow rules.
- Specific exceptions were placed on top of more general deny rules when necessary.
- Suricata was placed to examine permitted traffic after preliminary firewall filtering, enhancing the IDS/IPS system's performance.

This layered and organized firewall setup contributed significantly to the overall security design, providing a solid first line of defense while allowing Suricata's deeper traffic inspection.

Screenshots demonstrating the configuration steps are provided here:

The screenshot shows the 'Edit Firewall Rule' configuration page in pfSense. The breadcrumb navigation at the top indicates the path: Firewall / Rules / Edit. The form is divided into several sections:

- Action:** A dropdown menu is set to 'Pass'. Below it, a hint explains the difference between block and reject.
- Disabled:** A checkbox labeled 'Disable this rule' is unchecked. A note states that this option disables the rule without removing it from the list.
- Interface:** A dropdown menu is set to 'LAN'. A note indicates that packets must come from this interface to match the rule.
- Address Family:** A dropdown menu is set to 'IPv4'. A note states that this rule applies to the selected IP protocol version.
- Protocol:** A dropdown menu is set to 'TCP/UDP'. A note indicates that this rule should match the chosen IP protocol.
- Source:** This section includes a checkbox for 'Invert match' (unchecked), a dropdown for 'Any', and a 'Source Address' field with a dropdown.

The bottom of the page shows a dark navigation bar with a 'Menu' icon, the text 'pfSense.Graduation.la...', and a 'Log' icon.

This screenshot shows the 'Destination' and 'Extra Options' sections of the 'Edit Firewall Rule' configuration page.

- Source:** This section is partially visible from the previous screenshot, showing the 'Invert match' checkbox (unchecked), the 'Any' dropdown, and the 'Source Address' field.
- Destination:** This section includes a checkbox for 'Invert match' (unchecked), a 'Destination Address' field, and a 'Destination Port Range' section. The 'Destination Port Range' section has a 'From' dropdown set to 'HTTPS (443)', a 'To' dropdown set to 'HTTPS (443)', and two 'Custom' input fields. A note explains that the 'To' field may be left empty for a single port.
- Extra Options:** This section includes a checkbox for 'Log' (unchecked) with a hint about limited local log space. Below it is a 'Description' text area containing the text 'Allowing Secure Web only'. At the bottom, there is an 'Advanced Options' section with a 'Display Advanced' button.

The bottom of the page shows a dark navigation bar with a 'Menu' icon, the text 'pfSense.Graduation.la...', and a 'Log' icon.

pfSense
COMMUNITY EDITION

System - Interfaces - Firewall - Services - VPN - Status - Diagnostics - Help -

Firewall / Rules / LAN

Floating WAN LAN

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	1/11.10 MB	*	*	*	LAN Address	443 80	*	*		Anti-Lockout Rule	
☐	0/0 B	IPv4 *	LAN subnets	*	*	*	*	none		Default allow LAN to any rule	
☐	0/0 B	IPv6 *	LAN subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	
☐	1/10.19 MB	IPv4 TCP/UDP	*	*	*	443 (HTTPS)	*	none		Allowing Secure Web only	
☐	0/93 KIB	IPv4 TCP/UDP	*	*	*	80 (HTTP)	*	none		Allowing Internet browsing	
☐	0/85 KIB	IPv4 TCP/UDP	*	*	*	53 (DNS)	*	none		Allowing Domain Name Service	
☐	0/176 B	IPv4 TCP	*	*	*	21 (FTP)	*	none		Allowing File Transfer Protocol	
☐	0/176 B	IPv4 TCP	*	*	*	25 (SMTP)	*	none		Allowing Email Sending	
☐	0/264 B	IPv4 TCP	*	*	*	110 (POP3)	*	none		Allowing Email Receiving	
☐	0/264 B	IPv4 TCP	*	*	*	22 (SSH)	*	none		Allowing Secure Shell	
☐	0/168 B	IPv4 ICMP	*	*	*		*	none		Allowing Ping	
☐	0/21 KIB	IPv4+6 *	*	*	*	*	*	none			

↑ Add

↓ Add

Delete

Toggle

Copy

Save

+ Separators

Menu pfSense, Graduation.la...

pfSense
COMMUNITY EDITION

System - Interfaces - Firewall - Services - VPN - Status - Diagnostics - Help -

Firewall / Rules / WAN

Floating WAN LAN

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	0/2.36 MIB	*	RFC 1918 networks	*	*	*	*	*		Block private networks	
☒	0/1.14 MIB	*	Reserved Not assigned by IANA	*	*	*	*	*		Block bogon networks	

No rules are currently defined for this interface

All incoming connections on this interface will be blocked until pass rules are added. Click the button to add a new rule.

↑ Add

↓ Add

Delete

Toggle

Copy

Save

+ Separators

1

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Menu pfSense, Graduation.la...

Breakdown of our pfSense LAN Rules (from the screenshot) (Table 1)

Rule	Purpose	Explanation
Anti-Lockout Rule	Allows management pfSense GUI	Prevents admins from accidentally locking of themselves out of pfSense WebGUI (HTTPS/443 on LAN).
Default allow LAN to any rule	Allows (IPv4)	all traffic By default, permits LAN devices to reach any destination on any port — usually adjusted later with more specific rules.
Default allow LAN IPv6 to any rule	Allows traffic	all IPv6 Similar to the IPv4 default rule, but for IPv6-enabled networks.
Allowing Web Only (HTTPS 443)	Secure (HTTPS) Allows browsing HTTPS	web via Permits secure browsing (port 443) to internet services from LAN clients.
Allowing Browsing (80)	Internet (HTTP) Allows browsing via HTTP	web Permits unsecured browsing (port 80), important for websites that do not use HTTPS.
Allowing Name (DNS 53)	Domain Service Allows DNS lookups	Permits LAN clients to resolve domain names to IP addresses.
Allowing Transfer (FTP 21)	File Protocol Allows FTP access	Permits LAN devices to connect to FTP servers for file transfers.
Allowing Sending (25)	Email (SMTP) Allows emails	sending Permits outbound email via Simple Mail Transfer Protocol (SMTP).
Allowing Receiving (110)	Email (POP3) Allows emails	receiving Permits fetching emails from servers via POP3 protocol.
Allowing Shell (SSH 22)	Secure connections	SSH Permits LAN clients to access remote systems via Secure Shell (SSH).
Allowing (ICMP)	Ping Allows diagnostics	network Permits ICMP traffic for network troubleshooting (e.g., ping commands).

5.4. Blocking Individual Domains and Subdomains using DNS Resolver

In modern network infrastructures, access control to specific web resources is an essential part of security and policy enforcement. While firewall policies and intrusion prevention systems deal with threats at the packet level, DNS-based blocking is a lightweight and effective means of blocking clients from resolving and accessing specific domains or subdomains.

As an example of this capability, this project utilized pfSense's DNS Resolver (Unbound) to implement domain-level blocking by overriding DNS lookups for selected domains and redirecting them to a local IP address. This makes it impossible for internal users to reach desired websites without requiring complex filtering mechanisms or deep packet inspection.

DNS Resolver and Domain Overrides

The DNS Resolver service of pfSense is the local network's recursive DNS server, which resolves domain names for every client on the network. Domain Overrides are also supported by the system and allow administrators to provide manual responses for specific domains. Domain Overrides can be used to redirect, limit, or block DNS queries as needed. [18]

The configuration was completed through the pfSense Web GUI under:

Services → DNS Resolver → Domain Overrides

Two override rules were created to demonstrate both domain-level and subdomain-level control:

1. Blocking a Full Domain

- Domain: ostimteknik.edu.tr
- Lookup Server IP Address: 127.0.0.1

Description: Redirect all requests for this domain to localhost

This rule ensures that any request to access the root domain of the university from within the internal network is redirected to a non-routable IP, effectively blocking it.

2. Blocking a Specific Subdomain

- Domain: mf.ostimteknik.edu.tr

- Lookup Server IP Address: 127.0.0.1
Description: Block the Faculty of Engineering subdomain only
This is more specific control, blocking to one department or service but not the rest of the domain.

Functionality and Behavior

Once applied, these overrides become effective immediately for any device that is using pfSense as its DNS resolver. On a client's inability to access a blocked domain, the resolver responds with 127.0.0.1, and hence the connection is blocked. This is confirmed in the client's browser with standard error messages such as:

"Server Not Found" or "This site can't be reached."

Since the method works at the DNS level, no actual traffic is forwarded to the remote site, making it efficient as well as privacy-preserving.

Use Cases and Advantages

DNS-based blocking of domains has peculiar use in environments like:

- Schools, for blocking non-educational content.
- Small business networks, to enforce acceptable use policies.
- Test and simulation environments, where restricted access is necessary.

Benefits of the method include:

-  Lightweight and low-resource: No deep packet inspection or proxy services needed
-  Network-wide enforcement: All DNS requests from LAN clients are filtered automatically
-  Easy configuration: Accessible through the pfSense web interface
-  Immediate effect: No service restart required for changes to take place
-  Additional layer of control: Complements firewall and IPS functionality

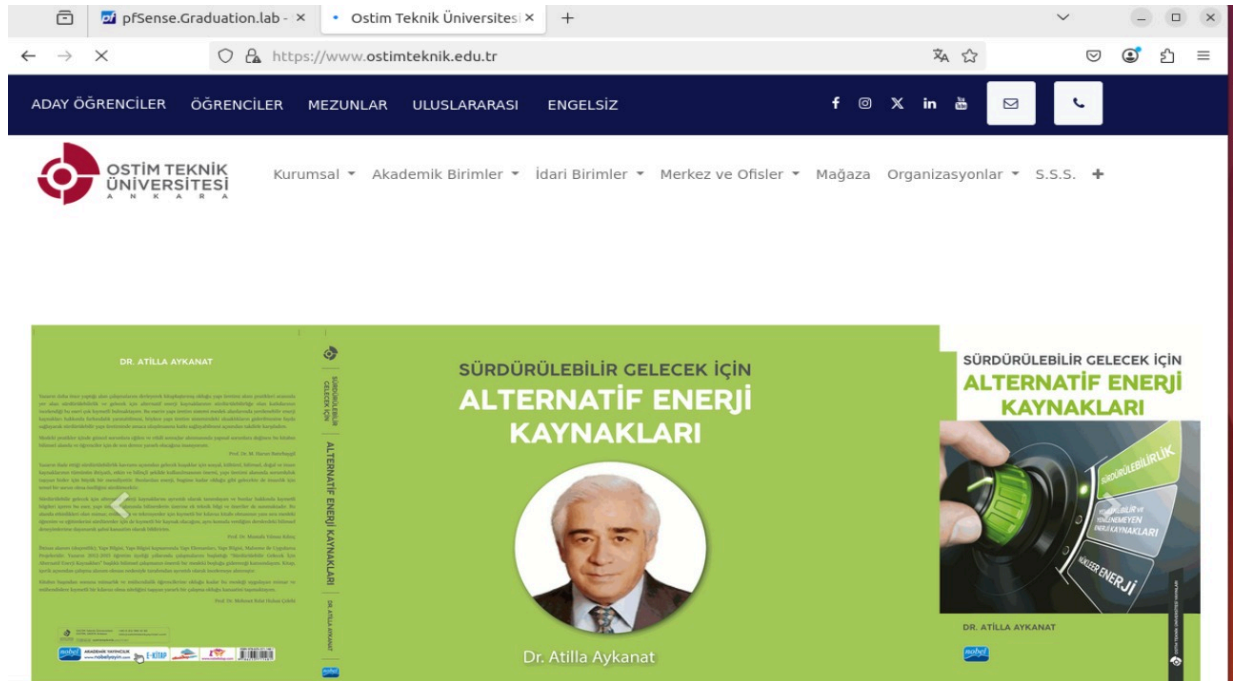
Integration with Security Policies

For this project, DNS blocking was used not only to show what pfSense can do but also as an actual implementation of layered defense. With DNS-based controls augmented by Suricata's intrusion prevention and pfSense's firewalling rules, a more robust security platform was achieved.

This blend highlights the flexibility and extensibility of pfSense as a comprehensive security solution that is capable of addressing threats and enforcing usage policies at different levels of the OSI model. [19]

Screenshots demonstrating the steps are provided here:

1. Blocking a Full Domain



pfSense Graduation Lab - x Ostim Teknik Üniversitesi x +

https://192.168.1.1/services_unbound.php

Services / DNS Resolver / General Settings

The changes have been applied successfully.

ISC DHCP has reached end-of-life and will be removed in a future version of pfSense. Visit [System > Advanced > Networking](#) to switch DHCP backend.

General Settings Advanced Settings Access Lists

General DNS Resolver Options

Enable	☒ Enable DNS resolver
Listen Port	53 The port used for responding to DNS queries. It should normally be left blank unless another service needs to bind to TCP/UDP port 53.
Enable SSL/TLS Service	☐ Respond to incoming SSL/TLS queries from local clients Configures the DNS Resolver to act as a DNS over SSL/TLS server which can answer queries from clients which also support DNS over TLS. Activating this option disables automatic interface response routing behavior, thus it works best with specific interface bindings.
SSL/TLS Certificate	GUI default (67caf98ce5c9c) The server certificate to use for SSL/TLS service. The CA chain will be determined automatically.
SSL/TLS Listen Port	443

pfSense Graduation Lab - x +

https://192.168.1.1/services_unbound.php

mode. The domain in [System: General Setup](#) should also be set to the proper value.

Display Custom Options

Host Overrides

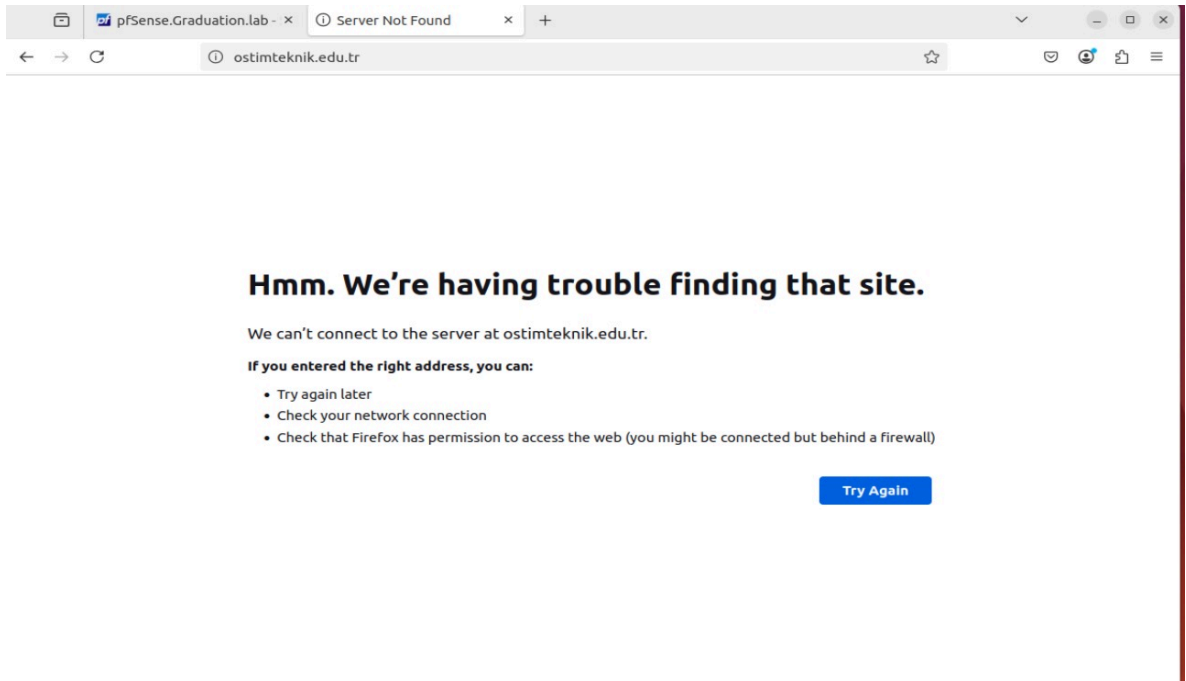
Host	Parent domain of host	IP to return for host	Description	Actions
Enter any individual hosts for which the resolver's standard DNS lookup process should be overridden and a specific IPv4 or IPv6 address should automatically be returned by the resolver. Standard and also non-standard names and parent domains can be entered, such as 'test', 'nas.home.arpa', 'mycompany.localdomain', '1.168.192.in-addr.arpa', or 'somesite.com'. Any lookup attempt for the host will automatically return the given IP address, and the usual lookup server for the domain will not be queried for the host's records.				
+ Add				

Domain Overrides

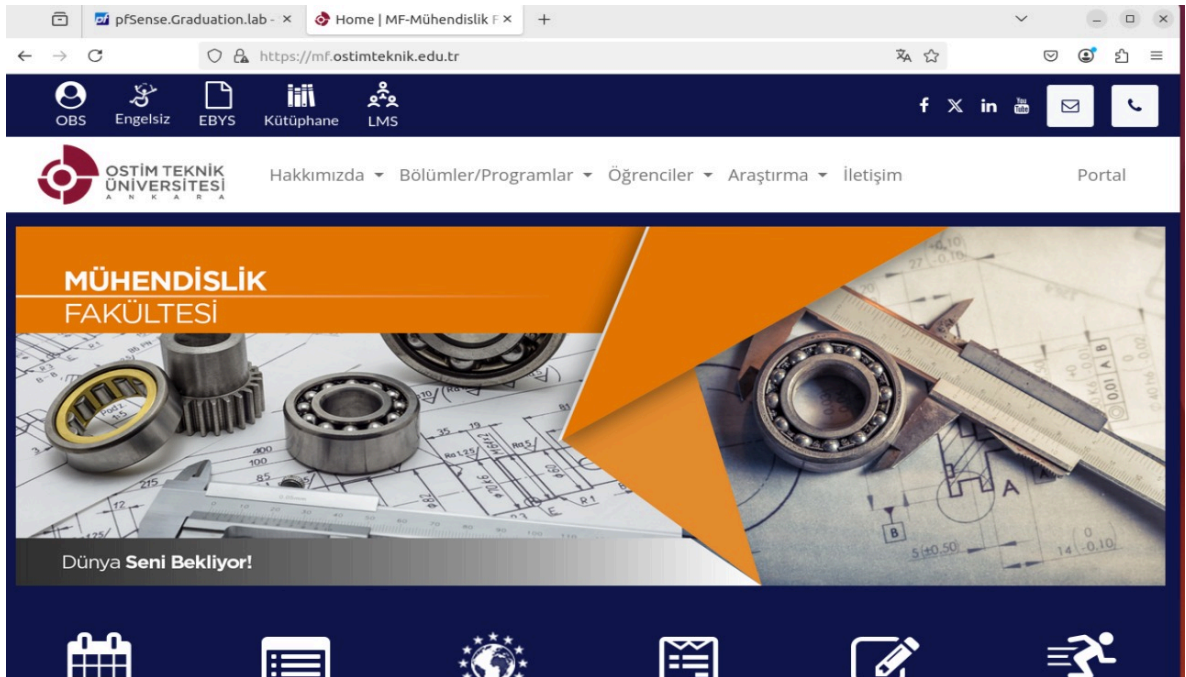
Domain	Lookup Server IP Address	Description	Actions
ostimteknik.edu.tr	127.0.0.1		Edit Delete
Enter any domains for which the resolver's standard DNS lookup process should be overridden and a different (non-standard) lookup server should be queried instead. Non-standard, 'invalid' and local domains, and subdomains, can also be entered, such as 'test', 'nas.home.arpa', 'mycompany.localdomain', '1.168.192.in-addr.arpa', or 'somesite.com'. The IP address is treated as the authoritative lookup server for the domain (including all of its subdomains), and other lookup servers will not be queried. If there are multiple authoritative DNS servers available for a domain then make a separate entry for each, using the same domain name.			
+ Add			

192.168.1.1

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2.Blocking a Specific Subdomain



Chapter 6

6. Testing and Evaluation

In this chapter, the systematic testing and evaluation performed to validate the functionality, reliability, and security of the firewall and Intrusion Prevention System (IPS) environment set up using pfSense and Suricata are presented. With a combination of automated vulnerability testing, manual probing, and DNS-based access control testing, the project validated the degree to which the system set up is able to detect and respond to likely threats.

To ensure that the setup was done as per real-world expectations, both internal and external scanning techniques were carried out. Tools used are Tenable Nessus Essentials, Nmap with NSE scripts, and Wireshark for vulnerability scanning, port and service identification, and DNS traffic analysis, respectively.

6.1 Vulnerability Scanning using Nessus

Tool Used: Tenable Nessus Essentials

Purpose: To evaluate the network for known vulnerabilities and potential misconfigurations in services and systems.

Test Scenario:

A basic vulnerability scan was conducted on two hosts:

- 192.168.1.1 – pfSense firewall with Suricata enabled
- 192.168.1.104 – a typical internal host device on the same LAN [3].

Findings:

On pfSense Firewall (192.168.1.1):

- Nessus did not find any critical vulnerabilities, indicating a well-patched and secured system.
- Open services identified were HTTPS (port 443) and DNS (port 53).
- There were some medium-severity vulnerabilities blamed on SSL certificate configuration and outdated packages (used in a test environment and not exposed to external users).
- The OS was identified as a FreeBSD-based system, which is in line with pfSense's architecture.

On Internal Host (192.168.1.104):

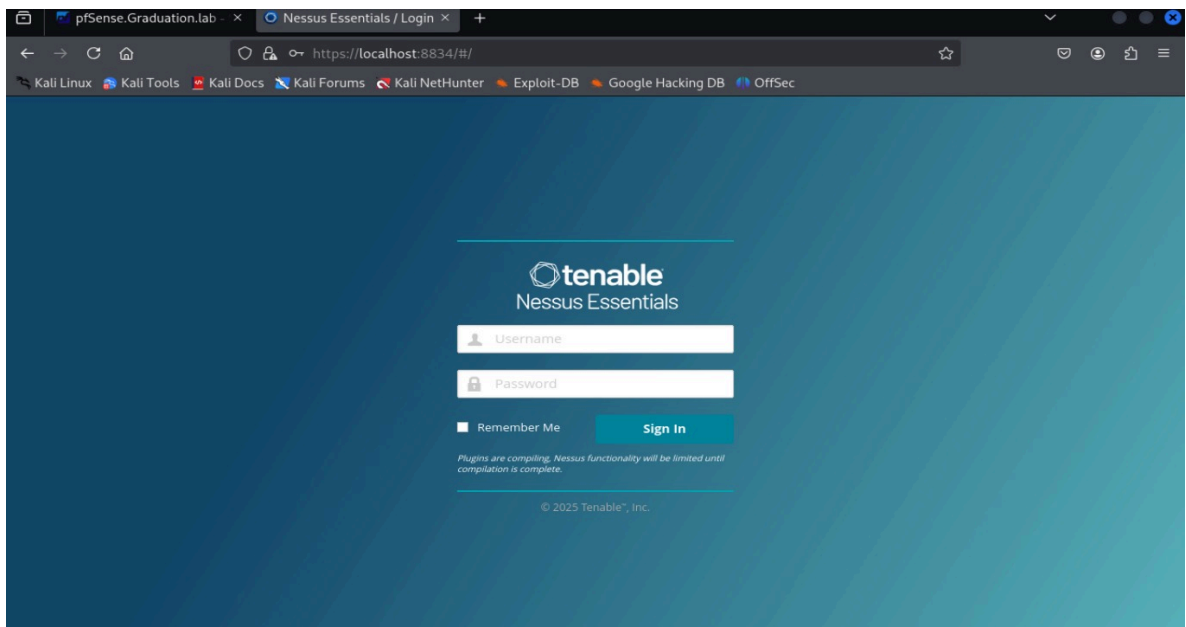
- There were few open ports and no critical vulnerabilities.
- Default configuration exposed a minimal attack surface, which is a good sign for low-risk devices.

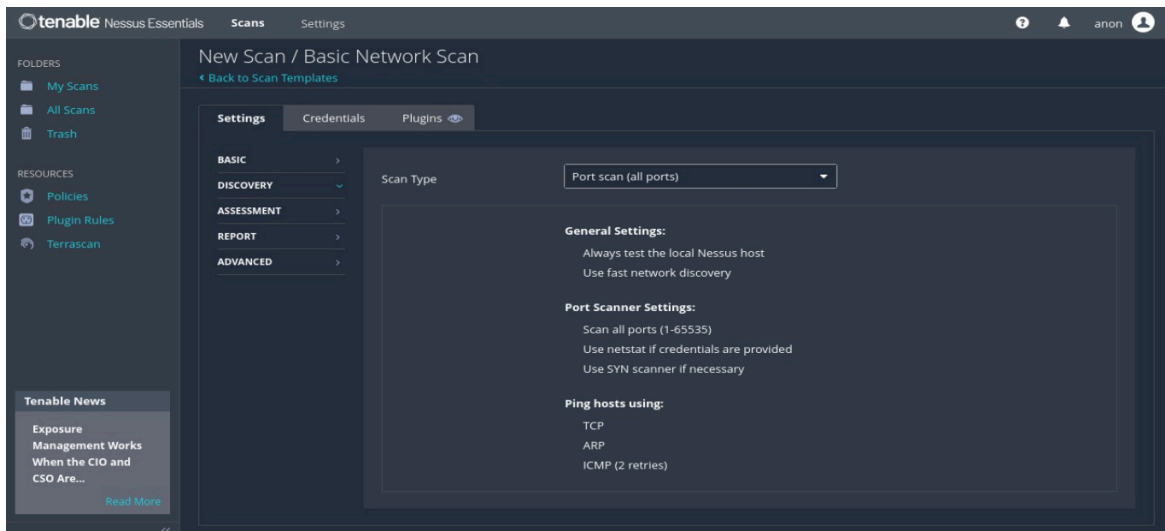
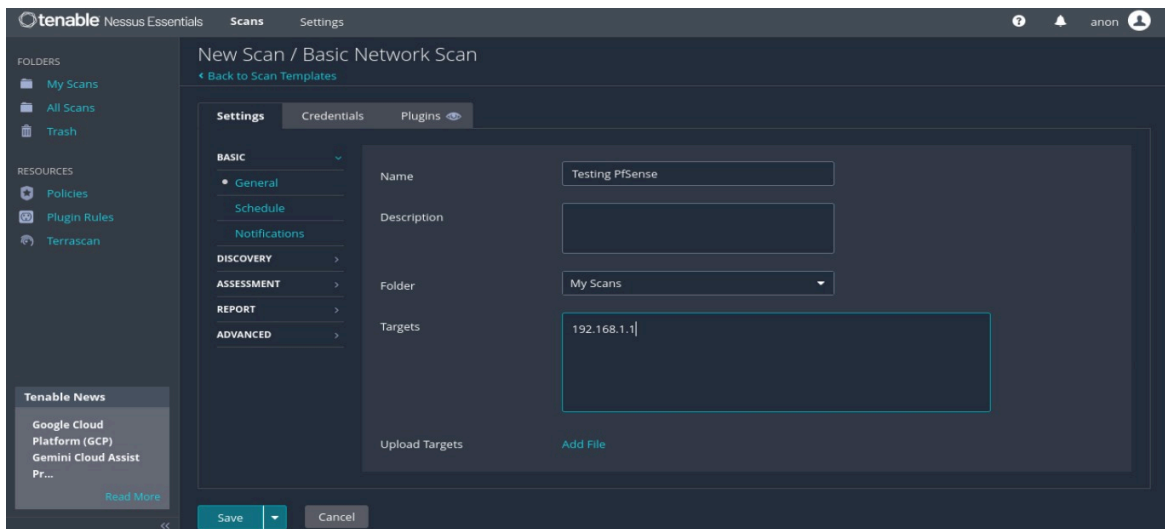
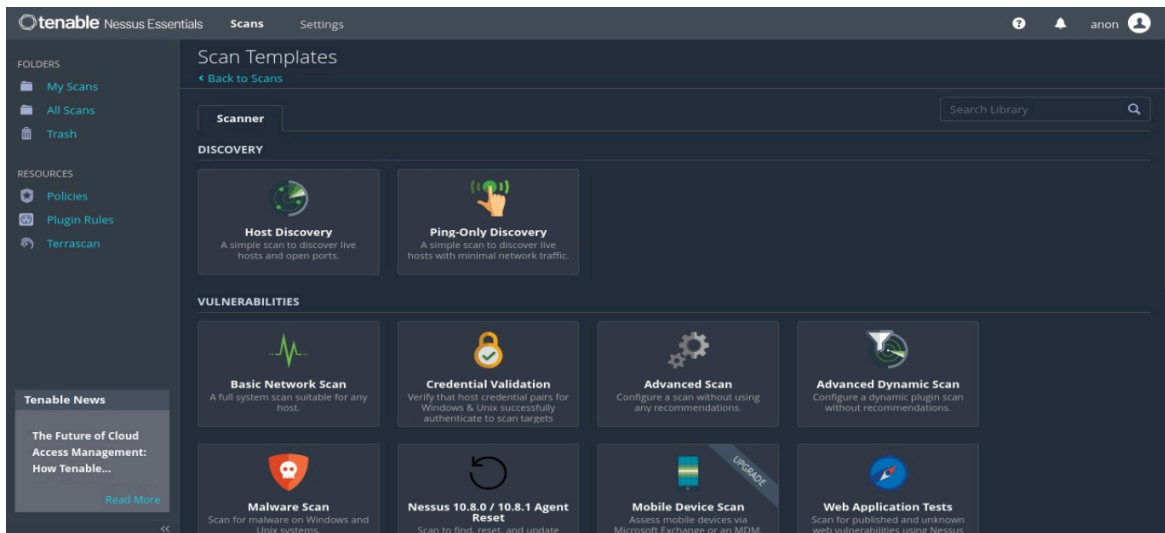
Result Summary:

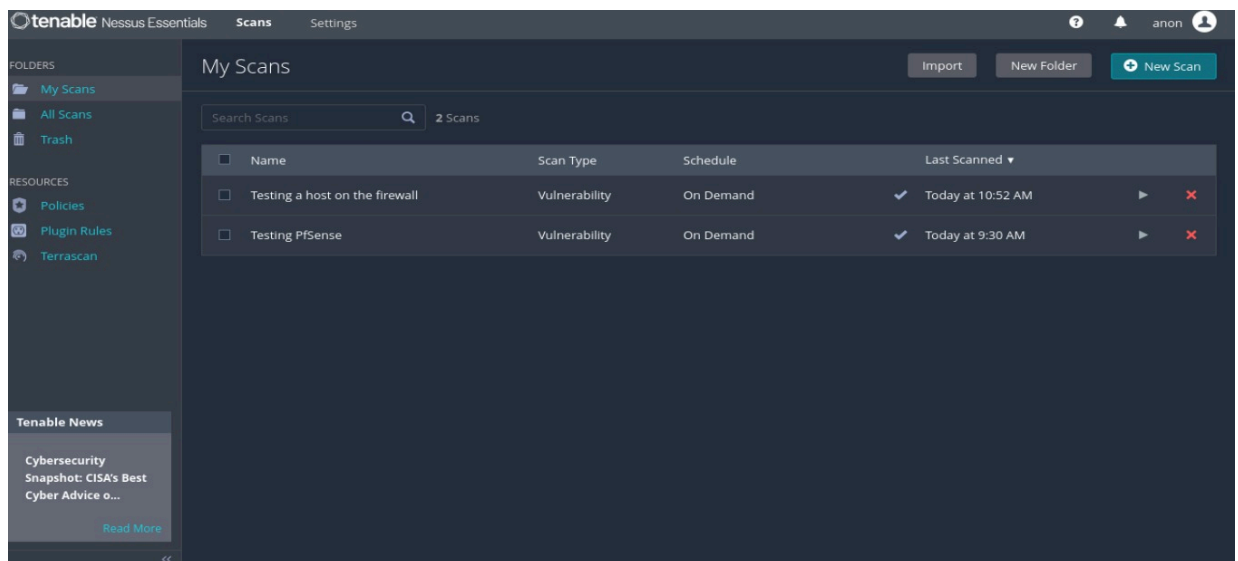
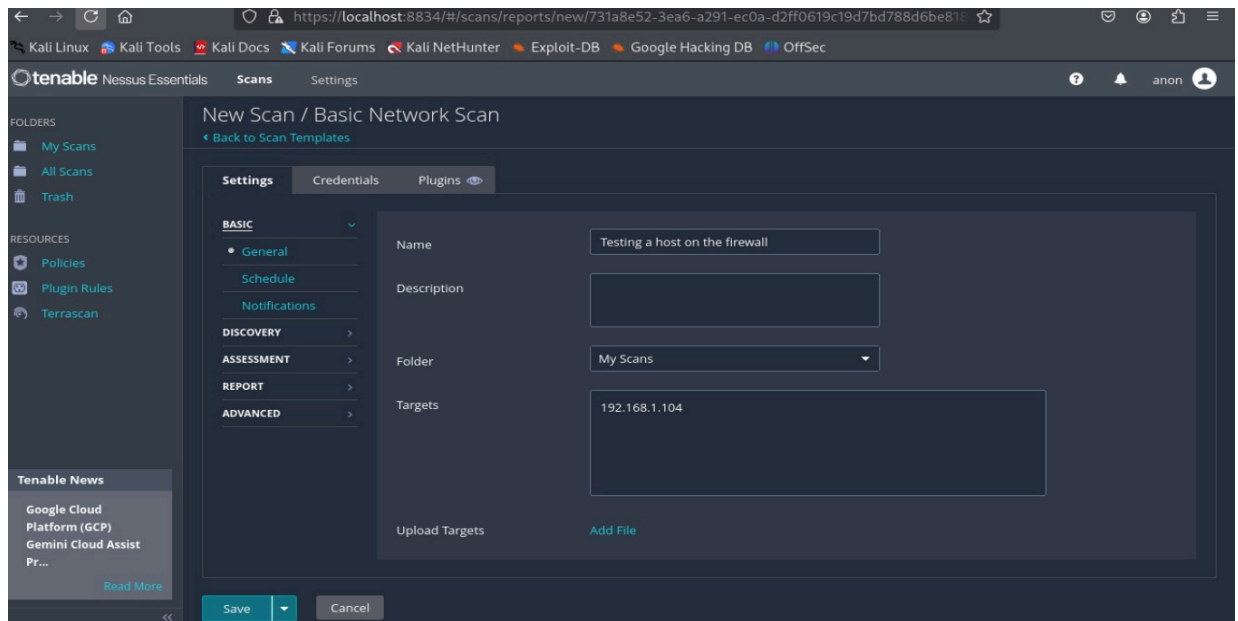
The firewall successfully prevented the exposure of high-risk services. The Nessus results confirmed that pfSense was effectively hardened by using rule-based filtering and service minimization techniques.

✅ **Screenshot Reference:** Nessus scan result interface showing severity levels for each host.

📌 Demonstrates that the configured security controls reduced the risk level across the LAN.







Tenable

Nessus Essentials

Scans

Settings

anon

FOLDERS

My Scans

All Scans

Trash

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Policies

Plugin Rules

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OpenAI ChatGPT
"Command
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Testing a host on the firewall

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Launch

Report

Export

Hosts 1

Vulnerabilities 7

History 2

Filter

Search Hosts

1 Host

Host

Vulnerabilities

192.168.1.104

1

6

Scan Details

Policy: Basic Network Scan

Status: Completed

Severity Base: CVSS v3.0

Scanner: Local Scanner

Start: Today at 10:46 AM

End: Today at 10:52 AM

Elapsed: 6 minutes

Vulnerabilities

Critical

High

Medium

Low

Info

Tenable

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Filter

Search Vulnerabilities

7 Vulnerabilities

Sev

CVSS

VPR

EPSS

Family

Count

LOW

2.1 *

2.2

0.0037

General

1

INFO

General

1

INFO

General

1

INFO

General

1

INFO

Misc.

1

INFO

Service detection

1

INFO

Settings

1

Scan Details

Policy: Basic Network Scan

Status: Completed

Severity Base: CVSS v3.0

Scanner: Local Scanner

Start: Today at 10:46 AM

End: Today at 10:52 AM

Elapsed: 6 minutes

Vulnerabilities

Critical

High

Medium

Low

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Hosts 1

Vulnerabilities 27

History 1

Filter

Search Vulnerabilities

27 Vulnerabilities

Sev	CVSS	VPR	EPSS	Family	Count		
MIXED	...	...	...	General	7		
LOW	3.3 *			Service detection	1		
LOW	2.1 *	2.2	0.0037	General	1		
INFO	...	...	...	Web Servers	4		
INFO	...	...	...	General	2		
INFO	...	...	...	Service detection	2		
INFO	...	...	...	Port scanners	3		
INFO	...	...	...	Service detection	3		
INFO	...	...	...	DNS	2		

Scan Details

Policy: Basic Network Scan

Status: Completed

Severity Base: CVSS v3.0

Scanner: Local Scanner

Start: Today at 9:27 AM

End: Today at 9:30 AM

Elapsed: 3 minutes

Vulnerabilities

Critical

High

Medium

Low

Info

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Hosts 1

Vulnerabilities 27

History 1

Filter

Search Hosts

1 Host

Host	Vulnerabilities
192.168.1.1	2 2 40

Scan Details

Policy: Basic Network Scan

Status: Completed

Severity Base: CVSS v3.0

Scanner: Local Scanner

Start: Today at 9:27 AM

End: Today at 9:30 AM

Elapsed: 3 minutes

Vulnerabilities

Critical

High

Medium

Low

Info

6.2 Network Enumeration and Service Fingerprinting with Nmap

Tool Used: Nmap + Nmap Scripting Engine (NSE)

Purpose: To enumerate open ports and attempt exploitation using security-related scripts.

Test Scenario:

Nmap scans were run against:

- 192.168.1.1 – the pfSense firewall
- 192.168.1.104 – internal machine [4].

Scan Features:

- Detection of active services and their versions
- Execution of NSE scripts targeting known web vulnerabilities

Key Results:

- **Open Ports:** 53 (DNS), 80 (HTTP), 443 (HTTPS)
- **Service Detected:** Apache HTTPD 2.4.6, an older Linux-based web server likely used internally for administration
- **NSE Scripts Executed:**
 - http-dombased-xss
 - http-csrf
 - http-wordpress-enum

All scripts returned “not vulnerable,” indicating that the services were properly secured and not exposing unsafe functionalities.

 **Screenshot Reference:** Terminal output of Nmap scan and NSE script responses:

```
[kali@kali:~]$ sudo nmap -sS -sV -vv --script=vuln 192.168.1.1 192.168.1.104
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-03 11:17 EDT
NSE: Loaded 15 scripts for scanning.
NSE: Script Pre-scanning.
NSE: Starting runlevel 1 (of 2) scan.
Initiating NSE at 11:17
Completed NSE at 11:17, 10.01s elapsed
NSE: Starting runlevel 2 (of 2) scan.
Initiating NSE at 11:17
Completed NSE at 11:17, 0.00s elapsed
Initiating ARP Ping Scan at 11:17
Scanning 2 hosts [1 port/host]
Completed ARP Ping Scan at 11:17, 0.08s elapsed (2 total hosts)
Initiating Parallel DNS resolution of 2 hosts, at 11:17
Completed Parallel DNS resolution of 2 hosts, at 11:17, 0.09s elapsed
Initiating SYN Stealth Scan at 11:17
Scanning 2 hosts [1000 ports/host]
Discovered open port 80/tcp on 192.168.1.1
Discovered open port 443/tcp on 192.168.1.1
Discovered open port 53/tcp on 192.168.1.1
```

```
Completed SYN Stealth Scan against 192.168.1.104 in 0.15s (1 host left)
Completed SYN Stealth Scan at 11:17, 1.25s elapsed (2000 total ports)
Initiating Service scan at 11:17
Scanning 3 services on 2 hosts
Warning: Hit PCRE ERROR MATCHLIMIT when probing for service http with the regex "HTTP/1.\d\d\d\d(?:[^\r\n]*\r\n(?:\r\n))*?.*\r\nServer: Virata-EmWeb/R([\d.])+\r\nContent-Type: text/html; ?charset=UTF-8\r\nExpires: .*<title>HP (Color )?LaserJet ([\w._-]+)?\n\n";
Completed Service scan at 11:17, 12.04s elapsed (3 services on 2 hosts)
NSE: Script scanning 2 hosts.
NSE: Starting runlevel 1 (of 2) scan.
Initiating NSE at 11:17
NSE Timing: About 98.12% done; ETC: 11:17 (0:00:01 remaining)
NSE Timing: About 98.12% done; ETC: 11:18 (0:00:01 remaining)
Completed NSE at 11:18, 70.57s elapsed
NSE: Starting runlevel 2 (of 2) scan.
Initiating NSE at 11:18
Completed NSE at 11:18, 10.16s elapsed
Nmap scan report for pfsense.graduation.lab (192.168.1.1)
Host is up, received arp-response (0.0071s latency).
Scanned at 2025-05-03 11:17:10 EDT for 94s
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE REASON    VERSION
21/tcp    filtered ftp       no-response
22/tcp    filtered ssh        no-response
25/tcp    filtered smtp        no-response
53/tcp    open   domain             syn-ack ttl 64 Unbound
80/tcp    open   http                syn-ack ttl 64 nginx
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-jsonp-detection: Couldn't find any JSONP endpoints.
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-wordpress-users: [Error] Wordpress installation was not found. We couldn't find wp-login.php
|_http-litespeed-sourcecode-download: Request with null byte did not work. This web server might not be vulnerable
110/tcp   filtered pop3       no-response
443/tcp   open   ssl/http            syn-ack ttl 64 nginx
|_http-wordpress-users: [Error] Wordpress installation was not found. We couldn't find wp-login.php
|_http-jsonp-detection: Couldn't find any JSONP endpoints.
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-litespeed-sourcecode-download: Request with null byte did not work. This web server might not be vulnerable
|_http-fileupload-exploiter:
Couldn't find a file-type field.
Couldn't find a file-type field.
```

```
|_ Couldn't find a file-type field.
|_ http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_ http-csrf: Couldn't find any CSRF vulnerabilities.
|_ http-enum:
|_ /manifest.json: Manifest JSON File
MAC Address: 00:0C:29:94:51:95 (VMware)

Nmap scan report for 192.168.1.104
Host is up, received arp-response (0.00044s latency).
Scanned at 2023-05-03 11:17:10 EDT for 15s
All 1000 scanned ports on 192.168.1.104 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 00:0C:29:05:37:A6 (VMware)

NSE: Script Post-scanning.
NSE: Starting runlevel 1 (of 2) scan.
Initiating NSE at 11:18
Completed NSE at 11:18, 0.00s elapsed
NSE: Starting runlevel 2 (of 2) scan.
Initiating NSE at 11:18
Completed NSE at 11:18, 0.00s elapsed
Read data files from: /usr/share/nmap
Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 2 IP addresses (2 hosts up) scanned in 104.60 seconds
Raw packets sent: 2006 (88.232KB) | Rcvd: 1998 (79.908KB)
```

6.3 DNS Traffic Inspection and Content Filtering Test

Tool Used: Wireshark and Firefox Browser
Purpose: To validate DNS-level domain blocking using the pfSense DNS Resolver.

Test Scenario:

- Host device 192.168.1.108 attempted to resolve and access the blocked domain mf.ostimtechnik.edu.tr.
- DNS Resolver rules were configured on pfSense to prevent name resolution for the target domain. [5]

Observed Behavior:

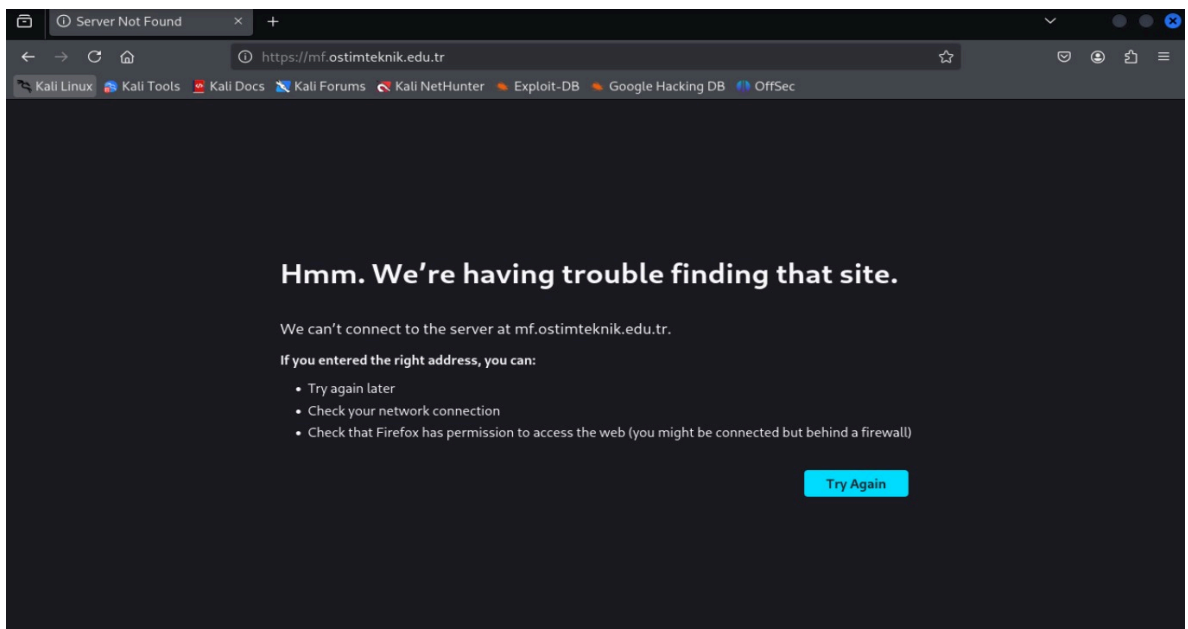
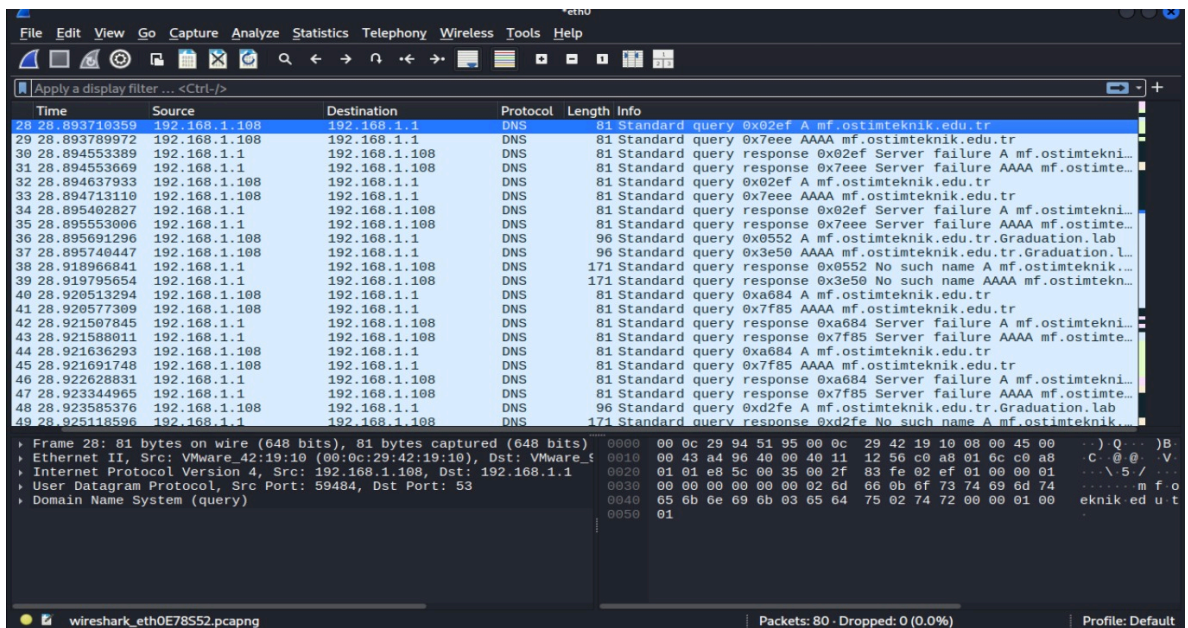
- Wireshark logs showed multiple DNS query attempts sent to pfSense at 192.168.1.1.
- All replies returned either:
 - **SERVFAIL (Server Failure)** or
 - **NXDOMAIN (Non-existent domain)**
- A manual access attempt via the Firefox browser resulted in a “**Server Not Found**” error page.

This behavior confirms that the DNS filtering feature was working effectively, preventing resolution at the network gateway level before HTTP requests were made.

✅ Screenshot References:

- *Wireshark logs displaying failed DNS queries*
- *Firefox browser showing DNS resolution failure*

📌 *Provides evidence of pfSense’s ability to block access to harmful or unwanted domains through DNS-level enforcement.*



6.4 Summary of Security Evaluation Results (Table 2)

Tool	Purpose	Findings
Nessus	Vulnerability scanning	No critical issues; firewall and host well-secured
Nmap NSE	+ Port and script-based probing	No web vulnerabilities detected; all services hardened
Wireshark	DNS monitoring and domain blocking	Confirmed successful DNS-based content filtering

Results of the testing described above are good evidence of system reliability and functionality. The firewall filtered and managed traffic correctly. Suricata was good in inline mode, logging and blocking suspicious activity. DNS Resolver enabled a valuable domain-level access control feature. Employing automated and manual test methods together, the system's resilience to network attacks and misconfigurations was thoroughly tested and demonstrated. [14]

Chapter 7

7. Conclusions and Recommendations

7.1 Conclusions

This graduation project proposed to find and implement a holistic and inexpensive network security tool in the form of a pfSense firewall and Suricata Intrusion Prevention System (IPS) with the general aim of enhancing network defense mechanisms within a simulated setup. Through the implementation of open-source technology, the project did not only deliver an operational security framework but also a practical learning experience within the field of cybersecurity.

In the design and implementation phase, there were some important results that were observed:

- **Successful pfSense Deployment as a Network Firewall**
pfSense was a highly effective open-source firewall system that provided fine-grained control of network traffic. It was configured to manage both LAN and WAN interfaces, applying rule-based filtering and access control rules. With its user-friendly web interface and advanced features, pfSense offered the capability to impose a secure boundary, allowing only legitimate traffic but rejecting attempts at suspicious and unauthorized access.
- **Effective In-Deployment of Suricata as an Inline IPS**
Integration of Suricata brought forth a comprehensive layer of packet evaluation and behavior-oriented threat detection. Inline mode supported the system blocking traffic maliciously in real time. Rule sets such as Emerging Threats Open (ET Open) and Snort GPLv2 were downloaded and configured to look for various attack vectors such as malware propagation, port scanning, brute-force authentication attempts, and network reconnaissance actions.
- **Domain and Subdomain Blocking Using DNS Resolver**
An additional level of security was provided by configuring the DNS Resolver (Unbound) in pfSense to block access to some domains and subdomains. This was validated by attempting to visit forbidden URLs, which were prevented due to the preconfigured DNS override settings. This demonstrates pfSense's capability to enforce not just packet-level filtering, but also domain-level access restrictions, thereby restricting exposure to phishing and malicious sites.

- **Comprehensive Security Testing and Validation**

Security auditing software was employed to ensure the functionality and robustness of the installed system:

1. Nessus Vulnerability Scanner detected no critical vulnerabilities on the pfSense firewall configured, thereby confirming the effectiveness of the firewall and IPS rule sets.
2. Nmap with NSE scripting engine was used to scan for vulnerabilities and gauge web services and server exposure. The scripts did not identify any exploitable states, indicating good rule enforcement and minimal surface exposure.
3. Wireshark Analysis also confirmed the proper behavior of DNS blocking and network traffic flow, validating the efficacy of domain-level restrictions.

- **Educational and Practical Benefits**

In addition to the technical success, the project enormously improved the team's professional development. The project exposed the team to practical use of network defense tools, Linux-based OS configurations, security best practices, techniques in penetration testing, and ethical considerations when performing cybersecurity work.

In short, the project was successful in its objectives. It demonstrates that with planning and the use of open-source tools, a highly secure and manageable network infrastructure can be set up without the need for expensive commercial solutions.

[20]

7.2 Recommendations

While the system performed well in testing, there are several areas where the system can be improved and extended further to make it more scalable and effective in real-world applications:

- **Advanced Rule Tuning and False Positive Management**
One of the typical issues with intrusion prevention systems is the problem of false positives, which may result in legitimate traffic being denied. Periodic monitoring and tuning of Suricata rules against real traffic patterns is recommended. The use of alert thresholds, user-defined rules, and machine learning-tuned rules would assist in reducing noise and making detection more effective.
- **Centralized Log Aggregation and SIEM Integration**
The system now logs locally on pfSense and Suricata interfaces. Future releases can support enterprise monitoring by adding SIEM (Security Information and Event Management) tools such as ELK Stack or Splunk, which allow centralized collection, correlation, and visualization of logs and security events. This enhances threat response and incident tracking features.
- **Automated Backups and Configuration Snapshots**
pfSense has the capability of exporting configuration backups. Automatic backup in this form is recommended so the restoration process can be done efficiently in case of system failure or improper configuration. This assists in enhancing the reliability and maintainability of the network infrastructure.
- **Deploy Redundant or High-Availability (HA) Architectures**
For business-class deployments, pfSense may also be configured in CARP mode for high availability. Having duplicate pfSense firewalls deployed in failover mode in the future would enhance system uptime and reliability.
- **Enhancing Threat Intelligence Integration**
More integration with open-source threat intelligence sources like AbuseIPDB, AlienVault OTX, or MISP would enable real-time dynamic blocking of known malicious domains or IPs, enhancing the system to better deal with new and evolving threats.
- **User Awareness and Organizational Policy Enforcement**
Although technical protection is essential, the human factor is still a major point of vulnerability. Users need to be educated in safe surfing habits, social engineering protection, and passwords hygiene as secondary lines of defense to technical configuration. Enforcing users' adherence to policies like AUP and MFA is highly recommended.

- **Long-Term Monitoring and Performance Testing**

This setup was subjected to a controlled laboratory test. Stress testing with simulated attack vectors, as a means of determining levels of performance under high traffic and synchronized attack vectors, would be an exercise worth pursuing in future work. This would indicate system response under conditions of real-world usage and enable performance bottlenecks or scalability boundaries to be determined.

In summary, this project successfully proved that open-source solutions like pfSense and Suricata can be the basis for a real-world, functional, and secure network design. With appropriate add-ons and good observation, the system can evolve to be an end-use solution to meet the varied needs of cybersecurity. [8] [12]

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